Renewable and Sustainable Energy Reviews 235 (2026) 116952

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

Water value and market response functions in hydropower-dominated electricity systems: A systematic review and methodological comparison

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HIGHLIGHTS

- Comprehensive review of water value methodologies across hydropower and economic models.
- Systematic literature screening and a classification framework of water value.
- Comparative analysis of LP, MILP, SDP, SDDP, and Econ-Engi modelling approaches.
- Stochastic models produce lower water values and wider spreads than deterministic ones.
- Research gaps identified for advancing integrated hydro-economic valuation.

ARTICLE INFO

Keywords:
Water value (shadow price)
Hydropower scheduling
Stochastic dual dynamic programming (SDDP)
Stochastic dynamic programming (SDP)
Linear/Mixed-integer programming (LP/MILP)
Economic models
Electricity market response functions
Meta-analysis

ABSTRACT

Hydropower operations increasingly rely on water value, the marginal opportunity cost of stored water, i.e., the shadow price of the reservoir balance, as a core signal for dispatch, bidding, planning, and multipurpose management. This systematic review (1931-2026; 277 studies) compares deterministic LPMILP, stochastic SDPSDDP, and economic-engineering approaches for estimating water value, harmonises terminology, and delineates water value from marginal production cost and market price. This article synthesises key determinants (inflows, storage and spill risk, prices, reserve commitments, head effects, and operational constraints) and analyses methodological trade-offs in uncertainty treatment, physical fidelity, and scalability. A meta-analysis, complemented by Wald tests, shows stochastic models yield lower central estimates and wider spreads than deterministic formulations, which reflects hedging under uncertainty. MILP often produces higher values when non-convex operational detail is represented. Water values are context-dependent: multipurpose and urban-supply reservoirs exhibit significantly higher values. Regional differences align with hydrological regimes and market design. This article discusses implications for operations and policy, including risk-aware, rolling-horizon scheduling; transparent reporting of assumptions (salvage value, scenario design, head modelling); co-optimisation of energy and ancillary services with attention to reserve asymmetries; and consistent treatment of pumped-storage arbitrage. This article highlights research needs in integrating environmental objectives under uncertainty, addressing non-stationary hydrology, coupling physical water values to financial risk management, and standardising reporting to improve reproducibility and comparability.

1. Introduction

Hydropower is widely recognised as a critical component of many electricity systems, providing both electricity generation and flexibility for balancing supply and demand [1,2]. Large reservoirs and Pumped-Storage Hydro (PSH) facilities enable the intertemporal shifting of energy production, linking present operational decisions with future opportunities and risks [3]. As the role of hydropower expands, understanding the economic value of stored water, commonly referred to as *water value*, is central to effective system operation and planning [4,5]. Accurate estimation of water value is critical for electricity market operation [6,7], long-term planning, and multipurpose

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https://doi.org/10.1016/j.rser.2026.116952
Received 3 November 2025; Received in revised form 24 March 2026; Accepted 25 March 2026
Available online 30 March 2026
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Nomenclature

Acronyms

CVaR Conditional Value-at-Risk
Econ–Engi Economic–Engineering Models
GVA Gross Value Added
KKT Karush–Kuhn–Tucker Conditions
LMP Locational Marginal Price
LP Linear Programming
MILP Mixed-Integer Linear Programming
PSH Pumped-Storage Hydro
SDDiP Stochastic Dual Dynamic Integer Programming
SDDP Stochastic Dual Dynamic Programming
SDP Stochastic Dynamic Programming

Sets

$\mathcal{D}$ Downstream reservoirs (in cascaded systems)
$\mathcal{E}$ Economic sectors
$\mathcal{F}$ Set of methodologies (modelling approaches in surveyed papers)
$\mathcal{G}$ Set of generators (hydropower, thermal, etc.)
$\mathcal{I}$ Water input types
$\mathcal{K}$ Set of break-points in piecewise-linear approximations (e.g., for head-discharge relationships)
$\mathcal{L}$ Set of locations (reservoirs, nodes)
$\mathcal{M}$ Set of regions (geographical areas in surveyed papers)
$\mathcal{O}$ Set of operation types (release, pumping, transfer, etc.)
$\mathcal{P}$ Set of purposes (water uses/demand categories)
$\mathcal{S}$ Set of storage levels (used in discretisation for SDP/SDDP)
$\mathcal{T}$ Set of time periods (e.g., hours, days, stages)
$\mathcal{U}$ Upstream reservoirs (in cascaded systems)

Parameters

$\bar{H}_t$ Effective hydraulic head at time t [m]
η_t^{cycle} Round-trip cycle efficiency for pumped-storage
η_t^{gen} Turbine efficiency at time t
η_t^{pump} Pumping efficiency at time t
η_t Turbine/generator efficiency at time t
SU^u Start-up cost of unit u [$]
$\overline{G}^u$ Maximum generation capacity of unit u [MW]
$\Phi(S_{T+1})$ Terminal (salvage) value function of storage at horizon T
ρ Density of water [kg/m^3]
GVA_j Gross Value Added of sector j
θ_{ij} Share of water input i in sector j
c^g Marginal cost of thermal generator g [$/MWh]
D_t System electricity demand at time t [MWh]
d_u Nominal water demand of use u [m^3]
g Gravitational acceleration [m/s^2]
p_t Electricity market price at time t [$/MWh]
$P_u(q)$ Marginal benefit function of water delivered to use u
$S^{\min}$, $S^{\max}$ Minimum and maximum reservoir storage [m^3]
Q Total available water resource [m^3]

Continuous Variables

λ_t^E Energy-based water value at time t [$/MWh]
λ_t^V Volume-based water value at time t [$/m^3]
λ_t Water value (shadow price of storage balance) at time t [$/m^3]
G_t^u Generation output of unit u at time t [MWh]
Sp_t Spill volume at time t [m^3]
E_t Evaporation/seepage loss at time t [m^3]
G_t^g Thermal generation of unit g at time t [MWh]
G_t Hydropower generation at time t [MWh]
I_t Natural inflow at time t [m^3]
P_t^{pump} Electrical input to pump at time t [MW]
Q_t^{npr} Non-power release (environmental/irrigation) [m^3]
Q_t^{pow} Power release volume at time t [m^3]
Q_t^{pump} Pumped inflow volume at time t [m^3]
$q_{t,u}$ Water delivered to demand/use u at time t [m^3]
R_t System reserves at time t [MWh]
S_t Reservoir storage at time t [m^3]
s_u Shortage for demand category u [m^3]

Binary Variables

u_t^u Commitment status of unit u at time t (1 = on, 0 = off)
y_t^u Start-up indicator for unit u at time t (1 if starting, 0 otherwise)

reservoir management, all of which guide dispatch, bidding strategies, and investment assessments [8,9].

Effective water management must balance energy generation with other essential non-energy uses, reflecting multiple objectives such as flood control, ecosystem preservation, and recreation [10,11]. These crucial constraints include meeting environmental and ecological requirements [12,13], such as maintaining downstream ecosystems (e.g., environmental flows or supporting the forest ecosystems), fulfilling needs for agricultural irrigation and human consumption, managing flood control, and supporting other uses like recreation [14,15]. Because these competing demands constrain the volume and timing of water available for power generation, they directly raise the marginal opportunity cost of stored water, the *water value*, which in turn determines how hydropower producers formulate their bids and supply functions in electricity markets [8,9]. Accurately capturing these multipurpose constraints within water value estimation is therefore essential not only for efficient reservoir operation but also for the design of market response functions that translate internal scarcity signals into credible price offers [6,7]. Despite the recognised importance of balancing these multiple water usages, the literature still encapsulates a large array of various modelling approaches with limited systematic comparison [16].

Even with the proliferation and methodological richness of these approaches, existing studies often differ in their conceptual rigor, computational tractability, and physical fidelity. These differences can lead to divergent water value estimates, with significant implications for market design, policy evaluation, and resource allocation [17,18]. A systematic review is therefore needed to clarify definitions, compare methodologies, and synthesize quantitative evidence across contexts. This review aims to:

(1) Survey methodological approaches for computing water value in hydropower-dominated systems.
(2) Compare these approaches based on their treatment of uncertainty, physical representation, scalability, and ability to inform market response functions.
(3) Synthesize quantitative evidence from the literature, highlight methodological trade-offs and identify research gaps for advancing water value estimation.

Several prior works have reviewed aspects of water value estimation and hydro-economic modelling. Existing systematic reviews of water-economy modelling applications focus on basin-scale economic models rather than electricity-market scheduling [16]. Similarly, surveys of optimal bidding strategies for hydro-electric producers concentrate on the strategic and stochastic dimensions of the bidding problem rather than on the water value concept itself [9]. The role of water value in electricity markets and hydro-thermal coordination has been examined

for the Spanish system [17]; that work was identified through our expanded search and is included in the reviewed corpus (Category A). While each of these contributions addresses important facets of the problem, none provides a unified, cross-methodological comparison of water value estimation spanning the full range of modelling traditions. The present review extends beyond these works in five specific ways: (i) it applies a PRISMA-based systematic protocol to a corpus of 277 studies published between 1931 and January 2026, ensuring transparency and reproducibility; (ii) it harmonises terminology across operations research, hydro-economics, and electricity-market design, clarifying how related terms (shadow price, scarcity rent, opportunity cost, dual value) map onto the same underlying concept; (iii) it presents a unified mathematical framework that covers Linear Programming (LP), Mixed Integer Linear Programming (MILP), Stochastic Dynamic Programming (SDP), Stochastic Dual Dynamic Programming (SDDP), and Economic-Engineering Models (Econ–Engi) models within a common notation; (iv) it conducts a quantitative meta-analysis, complemented by Wald tests, comparing water value estimates across methods, regions, and use purposes; and (v) it identifies the methodological drivers behind differing water value estimates, and shows that the choice of optimisation technique is statistically consequential for both the level and dispersion of reported values.

The remainder of the paper is organized as follows: Section 2 defines key concepts and terminology related to water value. Section 3 details the systematic review methodology. Section 4 presents the analytical framework and methodological comparison. Section 5 reports quantitative results and meta-analysis and discusses applications and policy implications. Section 6 outlines future research directions.

2. Conceptual framework and key definitions

This section establishes the conceptual foundation for water value in hydropower systems, clarifies its mathematical definition, distinguishes it from related concepts, identifies key determinants and drivers, and harmonises terminology used throughout the literature.

2.1. Terminology clarification

The term *water value* refers to the economic value that water acquires in hydropower operations when it is scarce relative to current and future needs. In the literature, it is also called marginal water value [4,19], opportunity cost of water [20–22], scarcity rent [23–25], shadow rent [26–28], or dual value/dual price [17,29,30]. Although these terms share a common conceptual core, they carry slightly different connotations across research communities: in operations research the preferred terms are *shadow price* or *dual value*, which represent the Lagrange multiplier on the storage balance constraint; in hydro-economics the literature favours *scarcity rent* or *scarcity value*, which emphasise the basin-wide or multi-sector allocation; and in electricity-market design the dominant terms are *water value* or *opportunity cost*, which refer to the bid-relevant marginal cost of hydropower generation. Despite these disciplinary nuances, all terms describe the same underlying concept: a dynamic measure of scarcity that emerges endogenously from optimisation or equilibrium models. For consistency, this paper uses “water value” throughout. A detailed mapping of these terms, their definitions, typical disciplinary usage, and representative references is provided in Table A.7 in the Appendix A.

2.2. Definition of water value

The concept of the *water value* provides the analytical backbone for decision-making in hydropower systems. It reflects the value of water primarily in terms of its potential for hydropower energy generation, though in multipurpose systems the value also incorporates competing non-energy uses [31–33]. While water stored in a reservoir has multiple non-energy uses, such as agricultural irrigation or human consumption, its value for energy production must be inferred indirectly through opportunity cost, as hydro resources typically have negligible short-term direct operation costs [34]. We define water value as the *marginal opportunity cost of stored water*, which represents the expected future income achievable from the hydropower energy that can be generated [6,35]. This marginal opportunity cost is the increase in the objective function (typically system cost minimisation or social welfare maximisation) associated with an incremental reduction in reservoir storage for energy production [34,36]. In optimisation terms, the water value corresponds to the Lagrange multiplier (or shadow price) on the reservoir balance constraint [30,37] and is used as the marginal production cost for the hydropower plants in dispatch models:

$$\lambda_t = \frac{\partial V^*}{\partial S_t}, \tag{1}$$

where V^* is the optimal value function and S_t is the reservoir storage at time t. The water value denotes the marginal value of stored water. In optimisation terms, it can be interpreted as the shadow price associated with the reservoir mass-balance (storage balance) constraint. Namely, the water value represents the change in the optimal objective value resulting from a marginal increase in water availability in the reservoir [4,17]. Water value may be expressed in volumetric terms ($/m^3) or, after conversion using hydraulic head and turbine efficiency, in energy-equivalent terms ($/MWh). Because it is a marginal quantity, a positive water value indicates that additional stored water is beneficial at the margin, whereas a value near zero indicates little marginal benefit. Water values are generally state- and time-dependent, varying across reservoirs, time periods, and hydrologic conditions [4].

Water value plays an important role in hydropower-dominated electricity systems. In short-term scheduling, it guides optimal hydro dispatch and bidding strategies in electricity markets [12,38,39]. In investment and planning contexts, water value supports long-term assessments of hydropower profitability [35,40], capacity adequacy [41,42], and coupling with other resources and storage [43]. In multipurpose reservoir systems, water value influences decisions that balance energy generation with irrigation [23,44,45], flood control [14,46,47], environmental flows [15,48,49], and navigation [50]. Because hydropower resources are often concentrated in regions with highly seasonal or uncertain inflows, the valuation of water under uncertainty is particularly critical for ensuring efficient, reliable, and sustainable system outcomes [4].

2.3. Distinction from related concepts (shadow price, marginal cost, market price)

It is important to distinguish water value from related but distinct notions:

(a) **Shadow price.** Water value is most precisely defined as the shadow price (Lagrange multiplier) on the reservoir balance constraint [51,52], in optimisation terms also known as the multiplier arising from the firms’ Karush–Kuhn–Tucker Conditions (KKT) [53]. It measures the marginal benefit of preserving an additional unit of water in storage for future use. However, models often generate multiple shadow prices simultaneously (e.g., on turbine capacity [54], spill constraints [55], environmental flow requirements [56], or transmission limits [57]). Care must therefore be taken to identify the multiplier associated specifically with the intertemporal storage balance Eq. (2), as this is the expression that corresponds to the water value [58,59]. Accounting or shadow prices on other constraints are important for water allocation and investment decisions but should not be conflated with the water value itself.

(b) **Marginal production cost.** The marginal cost of hydropower production at any given time is typically close to zero [33,39,60], since water is a fuel-free resource (apart from minor variable operation and maintenance costs) [61,62]. By contrast, water value reflects the *opportunity cost* of using water now rather than in the future. It is thus a measure of *scarcity* rather than a production cost [22]. In periods of abundant inflows, water value can fall to zero, while during scarcity it

can rise substantially, which aligns with the economics of other scarce resources [24].

(c) **Electricity market price.** In a competitive electricity market, the spot price p_t equals the short-run marginal cost of the last unit dispatched [63]. In hydro-dominated systems, this marginal cost can itself be the opportunity cost of water stored in the reservoir. Nevertheless, the water value λ_t and the market-clearing price remain analytically distinct. Specifically, p_t reflects the marginal *system-wide* cost of serving demand, whereas λ_t measures the *internal* opportunity cost of retaining water for future use [64]. In competitive, price-taking settings, optimal hydro bidding strategies therefore set offer prices close to the water value ($b_t \approx \lambda_t$) [65,66], translating internal scarcity into external price signals. Water value can also be inferred *ex post* from observed market prices and the variable production cost of the power system [67,68]. Yet, despite this operational link, p_t and λ_t capture different economic concepts and should not be conflated.

(d) **Price reaction, price response or supply function.** In equilibrium and agent-based formulations, the water value defines the producer's marginal cost curve, which serves as the basis for constructing supply functions or offer curves in electricity markets [15,61]. A water value curve, plotted against reservoir storage or release volume, represents the hydropower producer's supply curve: it determines the optimal production level by intersecting with the market demand function [18,38,69]. The slope of a residual demand curve can also indicate changes in profit in response to the producer's reserve offers [70,71], which relates to the producer's price-quantity response. While the marginal willingness to pay for alternative quantities of water can be interpreted as a demand curve [45], the water value curve discussed here represents the supply side: it reflects the producer's opportunity cost of releasing stored water. This supply-side usage extends beyond the optimisation-based definition and captures the strategic behaviour of hydropower firms in oligopolistic markets [29,72]. While related, it should be recognised as a behavioural interpretation rather than a purely technical or economic one.

2.4. Determinants and drivers (hydrology, market, operational, modelling)

Water value in hydropower systems is shaped by a range of physical, market, operational, and modelling factors. Throughout this section, water value λ_t is understood as the (shadow) marginal value of an incremental unit of storage in an economic optimisation whose primary objective is to maximise hydropower-related revenues; competing or non-energy uses enter either as additional terms in the objective or, more commonly, as operational constraints. Key determinants include:

(a) **[Physical] Hydrological Conditions:** Water values are highly sensitive to inflow variability and reservoir storage levels [73,74]. Typically, water value λ_t decreases as stored volume or inflow increases, due to the rising risk of spill and reduced scarcity [4,25,55]. However, in some cases, higher reservoir levels can increase water value when spill risk is absent and head effects dominate generation efficiency [12,35,75]. Because inflow conditions directly determine the scarcity rent embedded in the water value, they propagate into market outcomes through the bid prices that hydropower producers submit: in dry periods, higher water values translate into higher offer prices, shifting the supply curve upward and raising market-clearing prices [61,68].

(b) **[Physical] Spill Risk:** As reservoirs approach maximum water elevation, the probability of spill rises, reducing the expected marginal value of stored water. Thus, λ_t generally declines with increased spill risk [46,47,76]. In market terms, elevated spill risk incentivises producers to offer generation at lower prices to avoid wasting water, effectively shifting the supply curve downward and depressing spot prices [18].

(c) **[Market] Market Dynamics:** Expected electricity prices and co-optimised market products (e.g., energy, reserves [44,73]) directly influence water value [77,78]. Notably, commitments for down-regulation capacity often require maintaining higher base generation, increasing water burn and raising λ_t relative to energy-only schedules [50,58,75]. These market-driven components of water value feed directly into bidding strategies: producers adjust their offer curves to reflect not only energy revenues but also the opportunity costs of reserve provision [21,70].

(d) **[Operational] Operational Constraints:** Minimum environmental flows (ϕ) tend to increase water value by reducing available head and volume, especially in wet seasons [48,56]. Conversely, tighter ramping limits reduce operational flexibility and typically have a negative impact on water value, with effects most pronounced during low-inflow periods [64]. These constraints propagate into market prices by altering the shape and position of the hydropower supply function: binding environmental flow requirements raise the effective marginal cost of generation, while ramping limits reduce the producer's ability to capture short-lived price spikes [65,79].

(e) **[Physical/Modelling] Temporal and Spatial Scope:** Water values vary with seasonality and geography, which reflect differences in hydrological abundance and market design [4]. Finer temporal granularity (shorter time steps) and less aggregation (more detailed asset modelling) often yield higher water values by exposing local constraints and flexibility [80–82]. The choice of temporal and spatial resolution, therefore, affects the granularity of the resulting bid curves: hourly models capture intra-day arbitrage opportunities that daily or weekly models smooth over, leading to systematically different market outcomes [81,83].

(f) **[Modelling] Modelling Choices:** Deterministic models tend to overestimate water value by neglecting uncertainty, while stochastic formulations produce an *expected* water value, averaged across multiple inflow scenarios, which is usually lower and more variable [61,84,85]. The specification of terminal (salvage) value and the treatment of non-linear head effects also materially impact λ_t [40,59,86]. Since the water value serves as the implicit bid price for hydropower, these modelling choices directly affect the supply functions submitted to the market and, consequently, the resulting market-clearing prices [61,69].

(g) **[Multi-use] Multipurpose and Emerging Uses:** Competing water uses (irrigation, recreation, etc.) can significantly increase water value, especially under scarcity or high alternative value scenarios [23,87,88]. When such uses are modelled as hard constraints (e.g., irrigation release requirements), they reduce the effective storage available for power and thereby raise λ_t. Conversely, if their benefits are monetised within the same objective function, they alter rather than merely constrain the underlying opportunity cost [57,89]. In electricity markets, the higher water values associated with multipurpose reservoirs translate into higher offer prices, which can raise system-wide clearing prices in regions where such reservoirs constitute a significant share of supply [17,18].

Over the past five decades, different methodological families have emerged to compute water values. Most commonly, we distinguish linear or non-linear, static or dynamic, and deterministic or stochastic models [16]. Early work focused on deterministic optimisation models, such as LP and MILP, where water values are readily obtained as dual variables of the storage balance ([81,90,91]). However, deterministic formulations struggle to capture hydrological variability and future uncertainty. Stochastic approaches, such as SDP and SDDP [44,92,93], were subsequently developed to provide dynamically consistent valuations under uncertainty, albeit with different computational trade-offs. In parallel, economic equilibrium models have embedded water value within market-clearing formulations [18,23,94], while Econ–Engi and integrated frameworks have expanded the scope to include environmental and ecological dimensions [15,24,95]. More recently, Approximate Dynamic Programming (ADP), Reinforcement Learning (RL), and data-driven statistical methods have been applied to approximate or infer water values and related price-response functions from empirical data.

3. Methodology of the systematic review

This section describes the systematic approach used to identify, select, and synthesize literature on water value estimation in hydropower and electricity market studies. The methodology is designed to ensure transparency, reproducibility, and alignment with established best practices for systematic reviews [96–98].

3.1. Review protocol and search strategy

The review followed a structured protocol, established prior to the literature search, to maintain rigor and consistency. The protocol defined the research question, scope, eligibility criteria, and procedures for data extraction and synthesis.

Research question: How has the concept of water value been defined, operationalised, and applied in hydropower planning, market modelling, and reservoir management?

Scope: The review focused on peer-reviewed studies addressing water value in the context of hydropower systems, electricity markets, and multipurpose reservoir management.

Search strategy: A comprehensive search was conducted across major academic databases, including *Scopus*, *Web of Science*, *ASCE Library* and *IEEE Xplore*. Additional targeted searches were performed in *ScienceDirect*, *SpringerLink*, and *Google Scholar* to ensure broad coverage. Search strings combined key terms such as:

> ("hydro" OR "hydropower" AND "market" AND ("water value" OR "value of water" OR "shadow price" OR "shadow value" OR "scarcity value" OR "scarcity rent" OR "shadow rent" OR "dual value" OR "dual price" OR "opportunity cost" OR "marginal cost"))

The search was limited to publications from 1931 to January 2026, which reflects the period during which modern hydroeconomic and market optimisation frameworks have been developed. To maximise coverage, the search was conducted using multiple query variants. In addition to the primary query shown above, supplementary searches replaced "hydropower" with the broader term "hydro" and employed targeted strings such as "water value AND electricity markets." These variants were applied across all databases to capture studies that use alternative terminology for the same underlying concepts. The combined search and screening process yielded a final corpus of 277 studies across categories A–H and R (see Table 1 and Fig. 1).

3.2. Screening, eligibility, and data extraction

Screening process: The initial search results were imported into a reference management tool, and duplicate entries were removed. Screening was conducted in two stages:

Stage 1 - **Title and abstract screening:** Articles were assessed for relevance to water value in hydropower and electricity market contexts.

Stage 2 - **Full-text screening:** Remaining articles were reviewed in detail to determine eligibility based on predefined criteria.

Review team: All screening, eligibility assessment, and data extraction were performed by the authors. At each stage, two authors independently screened each record, with discrepancies resolved through discussion; if consensus could not be reached, a third author adjudicated. No external reviewers were engaged. While our dual-screening protocol aimed to minimise error, the absence of external reviewers remains a limitation, as independent validation could have offered a further check against potential confirmation bias.

Eligibility criteria: Studies were included if they were: (i) explicitly defined or applied the concept of water value (shadow price, opportunity cost, or equivalent). (ii) Focused on hydropower systems, reservoir operation, or engagement with electricity markets. (iii) Provided methodological detail on optimisation, modelling, or valuation frameworks.

Studies were not included if they were: (i) unrelated to hydropower or electricity markets (e.g., irrigation-only contexts), (ii) purely descriptive studies lacking analytical or optimisation components, and (iii) non-English publications, due to practical constraints.

Background references: In addition to the systematically screened studies (categories A–H), the review draws on a set of narrative references (category R) that provide methodological or contextual framing. Category R references support definitions, historical context, and cross-disciplinary connections throughout the manuscript but did not themselves meet the eligibility criteria for direct water value estimation or methodological comparison. They are therefore excluded from the quantitative data extraction and meta-analysis. The distinction between category R and the systematically screened categories is noted in Table 1.

Data extraction: A standardised data extraction form was developed to capture key information from each eligible study, including: (i) bibliographic details (authors, year, source), (ii) geographical and system context (region, type of hydropower system, market setting), (iii) conceptual framing of water value (definition, interpretation), (iv) methodological approach (optimisation technique, temporal and spatial scope), (v) key variables and parameters (inflows, storage, releases, market price), and (vi) reported outcomes (valuation metrics, uncertainty, policy implications). The complete screening trail, classification metadata, and extracted water value data points are publicly archived and documented in the companion dataset articles [99,100].

3.3. Categorisation and synthesis approach

Categorisation: Screened documents were classified using a structured framework (see Table 1), which distinguishes studies by their focus on hydropower production, market impacts, water value estimation, and methodological orientation. This categorisation enabled systematic comparison across varying modelling traditions.

Synthesis: The extracted data were synthesized using both qualitative and quantitative methods:

Qualitative synthesis: Studies were grouped by conceptualisation of water value, methodological approach, and application domain. Thematic analysis identified common trends, divergences, and gaps in the literature.

Quantitative mapping: Where possible, studies were aggregated by model family (e.g., LP, SDP, SDDP, equilibrium models), region, and use purpose. Reported water values and measures of uncertainty were compiled for meta-analysis.

A PRISMA-style flow diagram (Fig. 1) summarizes the screening and selection process, showing the number of records identified, screened, and included for synthesis.

Limitations: The review is subject to limitations, including database coverage, language restrictions, and the likelihood that studies reporting positive findings are over-represented. Even so, the rigorous protocol and transparent synthesis provide a robust foundation for the subsequent analysis of water value methodologies and outcomes.

3.4. Inductive and deductive methodologies in water value research

Scientific inquiry into water value estimation in hydropower systems can broadly employ either inductive or deductive methodologies [4]. Inductive approaches involve deriving general principles from specific observations, such as inferring water value relationships from empirical case studies [101,124,144], historical market data [66,68,102], or observed operational outcomes [69,83,151]. This usually involves econometric analysis of historical electricity prices and production of hydro reservoirs to identify key influencing factors and empirically estimate supply curves [69,83,164]. In contrast, deductive methodologies start from established theoretical frameworks, such as optimisation theory, economic principles, or physical laws, and apply these to derive specific, testable predictions about water value under various scenarios [244].

In this systematic review, we focus primarily on deductive methodologies. The rationale for this emphasis is twofold. First, deductive

Table 1
Designation of categories used in the screening process.

Category	Criteria	References
A	Hydropower production, water value, is directly related to the effect on the market when hydropower represents a large share of the electricity supply.	[7,16–20,32,33,37–42,45,46,54–57,62,63,66,67,70–73,78,81,84,86,89,91,94,101–131]
B	Hydropower production, water value, but not directly related to market impact.	[4,12,15,23,30,35,43,44,48,50,64,65,74,75,79,82,85,88,132–136]
C	Hydropower production and market, but not the water value.	[13,21,22,52,59,61,69,77,83,86,87,137–173]
D	Market, but not related to the shadow price and hydropower, with a large share of supply.	[51,58,80,158,174–189]
E	Market and shadow prices, but not hydropower, with a large share of supply.	[26–29,49,95,190–200]
F	Related, but presents an analysis rather than a methodology for estimating the effect of hydropower generation on markets or the water value.	[68,201–212]
G	Related, but the focus is not centered on the market, water value, or large hydropower share.	[11,36,60,90,123,213–243]
H	Water value, but not focused on market implications.	[14,24,244–256]
R*	Narrative references for background.	[6,9,10,34,257–287]

* Category R references provide narrative background and contextual framing; they are not part of the systematically screened corpus used for quantitative data extraction and meta-analysis.

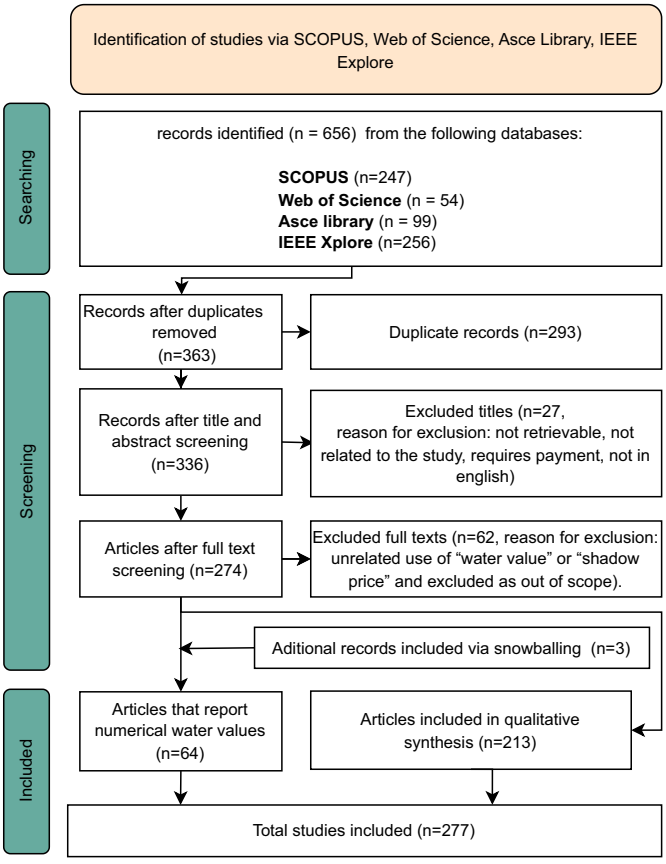


Fig. 1. Flow diagram of systematic review.

models, such as LP [29,67,108], SDP [56,61], and economic equilibrium frameworks [77,95], provide a transparent and reproducible basis for water value estimation [90]. They allow for explicit representation of system constraints, such as discharge limits [12,75,79], reservoir levels [88,120], production limits [30,39,91], water balance [4,37,55], and power flow capacities [57,59,94], objective functions, such as minimising operational costs or maximizing profit/user benefit [59,214], and uncertainty, particularly concerning future water inflow scenarios [16,61], which allows rigorous comparison across studies [252]. Second, deductive approaches are better suited to generalisation. By starting from first principles, they enable the synthesis of results that are not limited to a single empirical context but can be applied to different hydropower systems and market designs [12,244].

Inductive studies offer valuable insights, especially in uncovering emergent phenomena and context-specific behaviours [7], such as identifying key influencing factors for hydro reservoir supply curves [61]. However, their findings are often less transferable due to dependence on local conditions and data availability. For instance, some methods may not be applicable to every country with hydro reservoirs, and findings from one regional market may not directly translate to another [83,124]. Case studies often focus on specific regions or bilateral agreements [55,59]. Moreover, the literature on water value in hydropower is dominated by deductive modelling traditions [68,84], which reflects the sector's reliance on optimisation and economic theory for operational and planning decisions [44,89,106]. This dominance is evident in the widespread use of stochastic and deterministic optimisation approaches, including SDDP, for hydrothermal scheduling problems. Operational and planning decisions in the sector heavily rely on optimisation tools like the short-term hydropower optimisation [6,38,39,166], multi-market optimisation [86,115,116], bi-level optimisation for long-term planning [105,150], and optimisation models for revenue maximisation [30,52,101]. These decisions also incorporate economic concepts such as marginal costs [29,69,77], welfare maximisation [45,104,194], and electricity pricing strategies [54,136,188]. For these reasons, our review prioritizes deductive methodologies and acknowledges the complementary role of inductive research in informing model calibration and validation [244].

4. Analytical framework of water value in hydropower systems

This section presents a unified overview of the principal methodological families used to estimate *water value* in hydropower systems. The conceptual framework for water value estimation in hydropower systems is illustrated in Fig. 2. Each approach is described in terms of its mathematical formulation, treatment of uncertainty, physical fidelity, and interpretation of water value. Notation is harmonised throughout for clarity and comparability. A comparative summary table is provided at the end.

To facilitate systematic comparison, the review of methodologies in Sections 4.2–4.7 is structured around four consistent dimensions: (i) treatment of uncertainty, ranging from fully deterministic to explicitly stochastic formulations; (ii) physical fidelity, capturing how faithfully each approach represents hydraulic head,

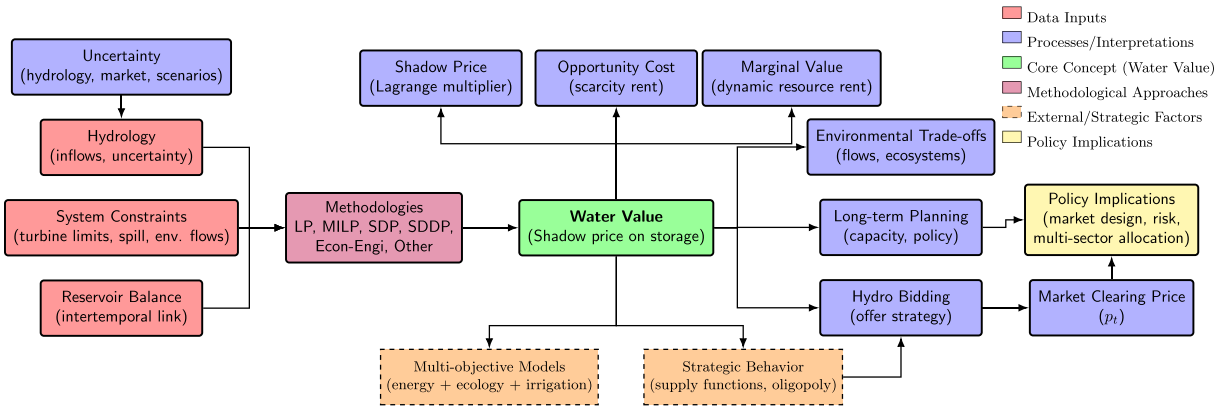


Fig. 2. Conceptual framework for water value estimation in hydropower-dominated electricity systems. The diagram illustrates the relationships among hydrological inputs, system constraints, methodological approaches, and the computation of water value (shadow price), as well as its interpretation and applications in market response, planning, and policy.

turbine efficiency, forbidden operating zones, and other non-linear hydro-technical constraints; (iii) representation of market design and market response, including the ability to model bidding, co-optimised products, and price formation; and (iv) computational scalability, reflecting the practical limits on system size, temporal resolution, and solution time. These four dimensions are summarised for all model families in Table 2. Section 4.11 and the preceding Section 4.12 then provide a cross-cutting synthesis discussing risk treatment, common pitfalls, and the practical implications of methodological choices for water value estimation.

4.1. Common reservoir dynamics and notation

All approaches are built on the fundamental reservoir mass balance:

$$S_{t+1} = S_t + (I_t + Q_t^{\text{pump}} - Q_t^{\text{pow}} - Q_t^{\text{npr}} - E_t)\Delta t, \tag{2}$$

where S_t is reservoir storage at time t, I_t is upstream inflow, Q_t^{pump} is pumped inflow, Q_t^{pow} is power release, Q_t^{npr} is non-power release (e.g., spillage, bypass, environmental flows), E_t is evaporation loss, and Δt is the time step. Storage is bounded: $S^{\min} \le S_t \le S^{\max}$.

Hydropower generation is given by:

$$G_t = \eta_t \rho g \bar{H}_t Q_t^{\text{pow}}, \tag{3}$$

where η_t is the turbine efficiency, ρ is the water density, g is the gravitational acceleration, and $\bar{H}_t$ is the effective hydraulic head. Water value λ_t is the shadow price (dual variable) associated with the storage balance constraint (2).

As reservoir storage S_t approaches its upper bound $S^{\max}$, the marginal probability of spill increases. This reduces the expected future benefit of stored water, causing the water value λ_t to decline [47,120]. Conversely, when spill risk is absent and head effects are significant, higher storage can increase λ_t due to improved generation efficiency.

4.2. Deterministic linear programming (LP)

In deterministic LP, all inputs (inflows, prices) are assumed to be known. The objective is typically to maximise profit or minimise system cost:

$$\max_{\{Q_t^{\text{pow}}, Q_t^{\text{npr}}, Q_t^{\text{pump}}, G_t^g, S_t\}} \sum_t \left(p_t G_t - \sum_{g \in \mathcal{G}} c^g G_t^g \right), \tag{4}$$

subject to reservoir balance (2), power conversion (3), storage bounds $S^{\min} \le S_t \le S^{\max}$, operational limits on releases, pumping and generation, and *linearised* version of the hydro–power conversion.

The physical energy Eq. (3) where $\bar{H}_t = f(S_t)$ is an (often affine) function of storage. Because (3) contains the product of two *decision variables* ($\bar{H}_t$ and Q_t^{pow}), it is **non-linear** and therefore inadmissible in a pure LP. Two standard remedies are:

Constant-head approximation: Fix $\bar{H}_t = \bar{H}^{\text{ref}}$ (e.g., average or maximum head). Eq. (3) reduces to the linear form $G_t = \kappa Q_t^{\text{pow}}$ with $\kappa = \eta \rho g \bar{H}^{\text{ref}}$.

McCormick (piecewise-linear) envelope [288] Introduce an auxiliary variable $z_t \approx \bar{H}_t Q_t^{\text{pow}}$ and add the four McCormick inequalities that bound z_t within the convex hull of the bilinear term over known bounds of $\bar{H}_t$ and Q_t^{pow}. The resulting formulation remains linear but approximates the true bilinear surface more accurately than the constant-head variant.

Water value: The dual variable λ_t associated with the balance constraint (2) represents the marginal value (*water value*) of an additional unit of storage at the beginning of period t.

Strengths: LP is computationally efficient, transparent, and scalable to large systems with deterministic inputs [108,123]. The resulting water values λ_t provide a useful deterministic benchmark for short-term operational planning. LP based economic dispatch has been applied across diverse market contexts [125,146,217].

Limitations: LP is *myopic* with respect to uncertainty; it assumes that future inflows and prices are perfectly known. *Linearisation error*: constant-head or envelope approximations may distort energy estimation and hence λ_t. In particular, constant-head approximations can introduce significant error in energy estimation and hence in the derived water value; McCormick envelopes improve accuracy but remain an outer approximation of the true bilinear surface. *Dual degeneracy*: multiple optimal dual solutions can yield non-unique or numerically unstable water values [106].

4.3. Mixed-integer linear programming (MILP)

A MILP formulation expands the deterministic LP of Section 4.2 with *integer* decision variables. Besides unit–commitment logic, the integer layer is also convenient for representing a variety of non-linear hydro–technical constraints such as forbidden (rough) operating zones, storage–to–elevation relationships and piecewise–linear head-dependent generation.

Let $\mathcal{T}$ be the scheduling horizon, $\mathcal{U}$ the set of thermal units, $\mathcal{R}$ the set of rough zones (forbidden discharge bands) and $\mathcal{K}$ the set of break-points in the piecewise-linear approximation of the head–discharge product. The short-run cost-minimisation problem can be written as

$$\min \sum_{t \in \mathcal{T}} \Big(\sum_{g \in \mathcal{G}} c^g G_t^g + \sum_{g \in \mathcal{G}} \text{SU}^g y_t^g \Big) \tag{5}$$

$$\text{s.t. } G_t^g \le \overline{G}^g u_t^g, \qquad \forall g,t \tag{6}$$

$$y_t^g \ge u_t^g - u_{t-1}^g, \qquad \forall g,t \tag{7}$$

Reservoir balance(2),

Operational limits on $Q_t^{\text{pow}}, Q_t^{\text{pump}}, S_t$,

Hydro–technical constraints (below).

$$u_t^g, y_t^g \in \{0,1\}, \quad \forall g,t. \tag{8}$$

Rough (forbidden) operating zones. Discharge bands that cause vibration or cavitation are ruled out via binary variables $z_t^r \in \{0,1\}$ [30,149]:

$$Q_t^{\text{pow}} \notin \left[\underline{Q}^r, \overline{Q}^r\right] \iff \begin{cases} Q_t^{\text{pow}} \le \underline{Q}^r - M(1-z_t^r) \\ Q_t^{\text{pow}} \ge \overline{Q}^r + M z_t^r \end{cases} \tag{9}$$

where M is a sufficiently large constant. The formulation forces the discharge to operate either below $\underline{Q}^r$ or above $\overline{Q}^r$.

Storage–to–elevation (head) function. The non-linear mapping $\bar{H}_t = f(S_t)$ is replaced by a piecewise-linear SOS2 approximation. An SOS2, or "Special Ordered Set of type 2", is a set of continuous variables with a special rule that at most two variables in the set can be nonzero, and if two are nonzero, they must be adjacent in a predefined order [289]:

$$\begin{aligned} &S_t = \sum_{k\in\mathcal{K}} s_k\,\alpha_{t,k}, \quad \bar{H}_t = \sum_{k\in\mathcal{K}} h_k\,\alpha_{t,k}, \\ &\sum_{k\in\mathcal{K}} \alpha_{t,k} = 1, \quad \alpha_{t,k} \in [0,1]\ \text{(SOS2)}, \end{aligned} \tag{10}$$

where (s_k, h_k) are the break-points of the storage-to-elevation function f.

Piecewise-linear hydropower conversion. The bilinear term $G_t = \eta_t \rho g\, \bar{H}_t Q_t^{\text{pow}}$ is linearised via a *lambda* formulation or a McCormick envelope: introduce $z_{t,k}$ s.t.

$$\begin{aligned} &Q_t^{\text{pow}} = \sum_{k\in\mathcal{K}} q_k\, z_{t,k}, \quad G_t = \sum_{k\in\mathcal{K}} \kappa_k\, z_{t,k}, \\ &\sum_{k\in\mathcal{K}} z_{t,k} = 1, \quad z_{t,k} \in [0,1]\ \text{(SOS2)}, \end{aligned} \tag{11}$$

with slopes $\kappa_k = \eta_t \rho g h_k$. If tighter accuracy is required, additional binaries may be used to enforce break-point selection, yielding a true MILP; otherwise, the SOS2 version remains a pure LP relaxation.

Water value: Because the presence of integer variables breaks the convexity of the feasible region, classical dual multipliers (shadow prices) are *not* uniquely defined. Approximate water values are usually obtained from the linear *relaxation* of (5)–(7) or via post–optimal sensitivity analyses that perturb the initial storage.

Strengths: MILP captures non-convex realities such as unit commitment [115,129], start-up costs [30,91,149], minimum generation levels [39,54], and rough zones [12,280]. It provides a flexible framework for embedding piecewise-linear approximations of storage–head and head–discharge relationships while keeping the model linear between break-points [37,120,290]. MILP formulations have also been extended to wind-hydro offering [228,273].

Limitations: Water values become *discontinuous* and harder to interpret relative to LP duals [54,156]. Computational burden grows rapidly with the number of time periods, break-points, and binary variables. Piecewise approximations introduce modelling error that must be balanced against tractability.

4.4. Stochastic dynamic programming (SDP)

Where the LP (Section 4.2) and MILP (Section 4.3) formulations optimise against a single deterministic scenario, SDP recognises that future inflows (and possibly prices) are random. The decision problem is solved stage-wise through the Bellman recursion:

$$V_t(S_t) = \max_{u_t} \left\{ \pi_t(S_t,u_t) + \mathbb{E}\left[V_{t+1}(S_{t+1}) \mid S_t, u_t\right] \right\}, \tag{12}$$

where S_t is the storage state, $u_t = (Q_t^{\text{pow}}, Q_t^{\text{pump}}, \dots)$ the control vector, π_t the single-period profit, and S_{t+1} follows the mass balance (2) with random inflow I_t. Practical implementations (i) discretise storage into a finite grid [64,85,88], (ii) pre-compute transition probabilities from historical inflows [36,74,172], and (iii) use piecewise-linear rewards so that head effects and rough-zone constraints [12,37,280] introduced in the MILP can be preserved without extra binaries.

Water value: $\lambda_t(S_t) = \frac{\partial V_t(S_t)}{\partial S_t} \approx \frac{V_t(S_t+\Delta S)-V_t(S_t)}{\Delta S}$., computed via finite differences.

Strengths: Captures uncertainty directly and therefore yields *risk-aware*, time-consistent release policies [93,214]. Non-linear head or turbine-efficiency curves can be embedded without linearisation error [120].

Limitations: "Curse of dimensionality" [167], where state and action spaces grow exponentially with the number of reservoirs and discretisation levels, restricting exact SDP to small systems or coarse grids [88]. Storage discretisation introduces aggregation error; finer grids improve accuracy but exacerbate computational cost [235]. Practical implementations for hydro scheduling typically discretise storage into tens to hundreds of levels per reservoir, but extending to multi-reservoir systems with more than three to four state variables becomes computationally prohibitive without aggregation or approximation techniques [85,167]. The choice of grid resolution, therefore, represents a fundamental accuracy/tractability trade-off: coarser discretisation speeds computation but increases interpolation error in the value function and hence in the derived water values [88,235].

4.5. Stochastic dual dynamic programming (SDDP)

SDDP bridges the gap between the single–scenario LP/MILP models (Sections 4.2 and 4.3) and the fully fledged SDP (Section 4.4). It keeps the underlying hydrothermal sub-problem *linear* (typically the same piecewise-linear head and rough-zone approximations used in the MILP) and iteratively constructs a multi-stage policy by adding Benders cuts.

$$V_t(S_t) \approx \max_{k\in\mathcal{K}_t} \{\alpha_t^k + \beta_t^k S_t\}, \tag{13}$$

where each affine cut (α_t^k, β_t^k) is generated in a *backward* pass from the dual of the linear sub-problem; a *forward* sampling pass then tests the policy on new inflow scenarios and refines the approximation.

Water value: At execution time, the marginal value of storage is given by the slope $\beta_t^{k\star}$ of the cut $k^\star$ that is active at S_t.

Strengths: Scales to long horizons and many interconnected reservoirs without the curse of dimensionality [55,141,260]. Inherits all linear/convex features of the LP formulation [89,92], and can accommodate risk measures (e.g., Conditional Value-at-Risk (CVaR)) with minor modifications [93,105,214]. Decentralized SDDP planning has been explored for large systems [103,263].

Limitations: The standard SDDP algorithm requires convex sub-problems; this is a core limitation that prevents the direct inclusion of integer decision variables such as unit-commitment binaries, start-up indicators, or non-convex hydraulic constraints. Integer decisions must therefore either be relaxed (e.g., via LP relaxation) or handled via specialised extensions such as Stochastic Dual Dynamic Integer Programming (SDDiP) [129,291]. This extension is discussed further in the Remark below. Physical accuracy is bounded by the linear approximations (head, efficiency, rough zones) adopted. Convergence speed and policy quality depend on the number and variety of generated cuts.

Remark: When non-convexities or integer decisions cannot be linearised, the method can be generalised into SDDiP. SDDiP solves mixed-integer sub-problems and generates *integer* Benders cuts [91,

129], enabling direct representation of unit commitment, start-up costs, or non-linear hydro functions [115,149]. Although more computationally demanding, SDDiP preserves the multi-stage stochastic structure of SDDP and therefore extends its applicability to problems that couple uncertainty with discrete hydro-thermal operating constraints [141,291].

4.6. Economic–Engineering (Econ–Engi) models

Econ–Engi models have been applied to transboundary water supply [191,192], conjunctive use [292,293], agricultural water value [246], production cost assessment [220], water pricing [193,251], and stochastic hydro-economic optimisation [196]. These models combine engineering constraints with economic objectives, often in a welfare optimisation framework:

$$\min_{x,q}\left[\underbrace{\sum_{t,p} C_{t,p}^{\text{scarcity}}(q_{t,p})}_{\text{economic scarcity costs}} + \underbrace{\sum_{t,l,o} C_{t,l,o}^{\text{op}}(x_{t,l,o})}_{\text{operating costs}}\right], \tag{14}$$

where $q_{t,p}$ is water delivered to use p at time t, and $x_{t,l,o}$ is water operation (release, pumping, etc.) at location l and time t.

Scarcity cost is defined as:

$$C_p^{\text{scarcity}}(q_p) = U_p(d_p) - U_p(q_p) = \int_{q_p}^{d_p} P_p(s)\,ds, \tag{15}$$

where $U_p(q_p)$ is the total benefit of delivery and $P_p(q)$ is the marginal benefit function.

Water value: The multiplier on the reservoir balance constraint represents both physical scarcity and economic rent.

Strengths: Captures multi-sector allocation and scarcity costs [41,42]. By embedding engineering constraints within an economic welfare framework, these models can simultaneously value water across competing uses (energy, irrigation, urban supply, environmental flows) and identify efficient allocation patterns [14,15].

Limitations: Requires demand curves and scarcity functions; calibration can be challenging, particularly when willingness-to-pay data are sparse or when transferring parameters across basins [28,117]. The data intensity and context-specificity of these models can limit their transferability to regions or sectors for which detailed economic data are unavailable [16].

4.7. Other methodological strands

Besides the LP/ MILP/ SDP/ SDDP and Econ–Engi families reviewed in detail, the literature also reports several alternative approaches [37].

Reinforcement learning and approximate dynamic programming algorithms learn release policies directly from data, handling high-dimensional, non-linear physics at the cost of opaque and noisy water value signals [9,186]. However, the learned policies are sensitive to reward function shaping, and the resulting water value signals can be opaque and noisy, making validation against benchmark solutions essential.

Meta-heuristic optimisers such as genetic algorithms, particle swarm optimisation, or simulated annealing treat the scheduling problem as a black box [91,281]. While easy to implement and capable of handling non-convexities, metaheuristics lack formal optimality guarantees and produce no dual information, which precludes direct extraction of water values.

Real-options models frame stored water as a portfolio of call options on electricity prices, providing analytical insight into time and volatility premiums while abstracting from detailed hydraulic constraints [8,9,46]. However, they require strong assumptions about price processes (e.g., geometric Brownian motion or mean-reverting dynamics) and abstract from detailed hydraulic constraints, limiting their applicability to operational scheduling.

Machine learning surrogates (regression trees, Gaussian processes, or neural networks) are trained on large libraries of pre-computed optimal solutions and deliver near-instantaneous water value estimates [138]. However, they suffer from extrapolation risk outside the training envelope and require large libraries of pre-computed optimal solutions for calibration, which limits their reliability in novel operating conditions or under non-stationary hydrology.

Agent-based and game-theoretic models focus on strategic bidding among multiple hydropower owners. Agent-based models have been developed for water markets [176,215,258,259]. Market simulation approaches include storage-specific models [216,222]. Water value emerges endogenously from Nash equilibria, but computational burden and behavioural assumptions limit their practical uptake [9,51,281].

Finally, Monte-Carlo simulation of fixed rule curves remains common in industry practice. Thousands of stochastic inflow–price traces are routed through detailed hydraulic simulators, after which the water value is inferred ex post from sensitivity tests. These methods are transparent and robust, yet inherently sub-optimal relative to truly optimised policies [128,294,295].

4.8. Summary of methodologies

Although methodologies reviewed from Sections 4.2–4.7 can yield estimates of water value, there are some nuances worth discussing in interpreting water values across different models. The fundamental differences in interpretation originate from different models' conceptual frameworks. For instance, deterministic models such as LP and MILP provide state-independent water values specific only to time period t [67,73]. In comparison, SDP and SDDP may generate water values that are state-dependent on the storage level both in the future and the past. Econ–Engi models generally yield state-independent water values as well, because researchers typically treat water resource quantity as an input to the model rather than modelling the dynamics of the storage level [14,15].

Regarding the capability of incorporating uncertainties into water value estimates, SDP, SDDP, and Econ–Engi models are superior to LP and MILP models. However, it is worth noting that uncertainty-capable models rely on researchers' assumptions about uncertainties, such as the probability distributions of inflows and energy prices [36,172]. To have reliable and valid results, the underlying assumptions on the uncertainties of SDP, SDDP, and Econ–Engi models require justification and calibration. We provide a summarised cross-comparison of the common methodologies employed in the literature in Table 2. For practical guidance on method selection under different operational conditions, see the decision guide in Table B.8 in the Appendix B.

4.9. Key analytical considerations and extensions

4.9.1. Units and conversions

Water values may be reported in volumetric terms ($/m^3) or energy terms ($/MWh). The conversion depends on head, efficiency, and time step:

$$\lambda_t^E = \lambda_t^V \times \frac{3.6\times 10^9}{\kappa_t}, \tag{16}$$

where $\kappa_t = \eta_t \rho g \bar{H}_t$ is the volume-to-energy conversion factor. The constant 3.6×10^9 is a unit conversion factor that arises from 3600 s/hr $\times$ 10^6 W/MW $= 3.6\times10^9$ W·s/MWh, converting from Joules (W·s) to Megawatt-hours (MWh). The factor κ_t has units of W/(m^3/s), or equivalently J/m^3, and represents the instantaneous power produced per unit volumetric flow rate at the given head and efficiency. Dimensional consistency can be verified as follows: λ_t^V in $/m^3 multiplied by the conversion factor in J/MWh and divided by κ_t in J/m^3 yields λ_t^E in $/MWh. Transparent reporting of units and conversion assumptions is essential for reproducibility.

Table 2
Comparison of methodological approaches to hydropower water value estimation.

Method	Uncertainty treatment	Physical fidelity	Scalability	Water value interpretation	Limitations	Reference
LP	None (deterministic)	Linear, simplified	High	Shadow price of storage	Ignores uncertainty, non-linearities	[10,26,38,56–59,62,67,87,105,106,108,111,115,123,125,146,154,155,162,175,180,181,184,187,206,208,210,211,263,270,282]
MILP	None (deterministic)	Linear, integer constraints	Medium	Approximate (LP relaxation)	Discontinuous, less interpretable	[30,39,40,42,48,54,65,70,73,78,80,86,91,114,128,138,144,149,150,158,273]
SDP	Explicit stochastic states	Non-linear, detailed	Low	Slope of value function	Curse of dimensionality	[12,33,40,56,64,70,71,71,79,84,85,89,110,122,123,127,189,231,283,285]
SDDP	Scenario-based stochastic	Linear, multireservoir	High	Slope of cutting plane	Linear physics, cut selection	[4,11,44,55,75,101,107,124]
Econ– Engi	Scenario or deterministic	Simplified, engineering + economic	Medium	Scarcity rent + physical shadow price	Data/ calibration intensive	[14,15,18,23,24,36,41,42,45,46,52,72,77,88,90,95,110,117,139,140,142,143,179,197,219,220,229,230,244,250,252,264,268,281,292,295]
Other models	Methodology specific	Simplified	Low-High	Post-ex	Water value not part of optimisation	[6,9,13,16,22,29,32,34,43,50,51,61,68,69,74,81–83,94,112,114,116,120,138,139,151,156–158,161,163–165,176,177,183,185,205,209,224,225,233,235,236,249,253,262,271,272,275,279,284,292,293,295–298]

4.9.2. Price and product mapping

Water value λ_t serves as a crucial input for informing market offers and bidding strategies for hydropower units [51,73,86]. For price-taking bids, a common approach is to approximately relate the bid to the expected energy value $b_t \approx \lambda_t^E$. Hydropower plants often participate as price-takers in day-ahead electricity markets [12,157]. The objective function in such scenarios, particularly in short-term scheduling, typically includes revenue obtained from the day-ahead market (hourly energy price multiplied by hourly generated power) [64,83]. Bidding strategies have been extended to profile block bids [160], multi-regional markets [168], reactive power pricing [170], sequential market clearing [232], coordinated bidding [189,199], demand response [264,270], and offer curve construction [132,148]. Similarly, the unitary price of pumping storage units is calculated based on the day-ahead market price, with any MWh added or absorbed for system balancing, incurring a penalty via the shadow price [80,175]. For co-optimised products (e.g., reserves), opportunity costs must account for activation probabilities and product-specific constraints. The overall objective function in short-term optimal operation models for hydropower reservoirs can include maximizing economic benefits from supplying both energy and spinning reserve [50,58]. The availability of spinning reserve is subject to minimum reserve constraints for buses and can be influenced by energy demand or grid security. Research also encourages incorporating alternative markets, such as the day-ahead automatic frequency restoration reserve market, into water value studies [21,114,187]. The modelling of PSH configurations within a Day-Ahead Unit Commitment (UC) model can include unit start-up and shut-down costs, as well as losses from maintaining spinning reserve units [43,78]. These formulations involve constraints on minimum and maximum power outputs, pumped power, turbine power, and unit status variables.

4.9.3. Reserve product asymmetry

Providing *down*-regulation capacity often introduces an asymmetry in both water use and opportunity costs for hydropower units. When the requirement for available down-regulation is *binding*, additional *must-run* generation is forced, which raises the shadow price of the storage balance (the water value λ_t) [40,211]. For PSH plants this can mean operating the pumps during typically low-price hours; the energy consumed could otherwise be sold, so the effective water burn and the opportunity cost increase [206,299]. Conversely, a binding *up*-regulation requirement creates head-room, reduces the average discharge, and therefore lowers λ_t [40]. While some studies assume symmetric reserve requirements, this assumption can be relaxed by introducing separate upward ($R_t^{\uparrow}$) and downward ($R_t^{\downarrow}$) reserve variables.

For generation schedule forced by reserve, let G_t^{base} be the scheduled (pre-dispatch) turbine output, $\tilde{G}_t$ the purely energy-oriented part of that output, and $\bar{G}_t$ the physical turbine limit. Reserve commitments impose

$$G_t^{\text{base}} = R_t^{\downarrow} + \tilde{G}_t, \qquad \text{(must–run for down reserve)} \tag{17a}$$

$$G_t^{\text{base}} + R_t^{\uparrow} \le \bar{G}_t, \qquad \text{(head-room for up reserve)} \tag{17b}$$

with dual variables $\mu_t^{\downarrow} = \lambda_{(17a)}$ and $\mu_t^{\uparrow} = \lambda_{(17b)}$, respectively. These multipliers feed into the reservoir balance so that the *effective* water value is

$$\lambda_t = \lambda_t^{\text{energy}} + \mu_t^{\downarrow} - \mu_t^{\uparrow}. \tag{18}$$

Hence, $\mu_t^{\downarrow} > 0$ (binding down reserve) raises λ_t, while $\mu_t^{\uparrow} > 0$ (binding up reserve) lowers it.

For PSH arbitrage conditions let P_t^{pump} be the electrical input and Q_t^{pump} the associated water volume pumped during time step t:

$$Q_t^{\text{pump}} = \frac{\eta_t^{\text{pump}} P_t^{\text{pump}}}{\rho g \bar{H}_t^{\text{pump}}}. \tag{19}$$

Define the round-trip efficiency factor

$$\eta_t^{\text{cycle}} := \frac{\eta_t^{\text{pump}} \eta_t^{\text{gen}} \bar{H}_t^{\text{pump}}}{\bar{H}_t^{\text{gen}}}. \tag{20}$$

After converting the volumetric shadow price λ_t [\$/m^3] to its energy equivalent

$$\lambda_t^E = \lambda_t \frac{\rho g \bar{H}_t^{\text{gen}}}{\eta_t^{\text{gen}}} \qquad [\$/\text{MWh}], \tag{21}$$

the no-arbitrage band for a loss-making cycle reads

$$\lambda_{t+1}^E \ge \frac{p_t}{\eta_t^{\text{cycle}}}, \qquad \text{(buy and pump if profitable)} \tag{22a}$$

$$\lambda_t^E \le p_t \eta_t^{\text{cycle}}, \qquad \text{(generate if profitable)} \tag{22b}$$

where p_t is the spot price in \$/MWh. Inequalities (22) emphasise that cycle losses create a band where operators pump only when the expected future energy value of stored water, λ_{t+1}^E, exceeds the discounted cost of pumping (22a), and generate only when the current energy value is below the spot price adjusted for efficiency losses (22b). Reserve provision studies include wind power reserves [152,239], PSH flexibility [282], and reserve exchange benefits [202].

4.10. Network and unit-commitment coupling

Coupling unit commitment and hydropower scheduling with network constraints such as the Direct Current (DC)/Alternate Current (AC)-Optimal Power Flow (OPF) reshapes marginal water values through locational prices and commitment-driven revenues, making consistent translation across temporal scales and asset aggregation essential for comparability and internal consistency [89,112]. Network coupling studies include LMP analysis [155,219], tunnel network modelling [236], flow mixing [249], decomposition approaches [296], and multi-provincial scheduling [279]. Practical implementations decompose horizons (short-, medium-, long-term) and often use master–slave structures, disaggregating weekly inflows and water values to hourly signals for bidding while aggregating smaller assets (including "united generators") to control dimensionality; these choices require careful parameterisation because turbine heterogeneity, hydro production function discretisation, and head-dependent non-convexities can otherwise distort value signals [62]. Power conversion is most often linearised with piecewise-linear *Q–H* envelopes, yet quadratic or bilinear formulations remain an alternative when solvers can handle convex (or mixed-integer convex) constraints, and offer higher fidelity without an explosion in break-points. In cascaded rivers, travel time is another source of coupling: multi-day time steps or delay equations are employed, but assuming a single deterministic lag neglects the fact that individual water parcels exhibit a distribution of travel times due to routing and forebay dynamics. Representing this dispersion, e.g., with lag distributions or storage-delay tanks, improves downstream head and spill estimates. To mitigate the curse of dimensionality and ensure accurate translation of results from regional or aggregated models to plant-level decisions, it is preferable to use system-wide or empirically derived pseudo-turbine supply curves rather than single-plant bidding in cascades [82,167], and to align risk horizons with long-term obligations. Finally, deterministic LP formulations and inconsistencies between short- and medium-term models can yield misleading recommendations due to non-anticipativity; rigorous cross-scale consistency is therefore critical for sound estimation of water values and financial risk [61,83].

4.11. Risk, robustness, and ambiguity

The definition of *optimal* in reservoir operations is inherently linked to the decision maker's attitude towards risk [252]. Risk-averse objectives (e.g., mean-risk, CVaR) and robust optimisation approaches typically yield higher water values under scarcity [81,105,113]. Reporting confidence intervals and performing out-of-sample validation [89], and sensitivity to risk parameters is recommended [33]. The aggregation of reservoirs can underestimate local water scarcity, potentially leading to erroneously higher water values in drier basins [260]. For hydropower operations, particularly when hedging against energy shortfalls, informed decisions require a careful balance between the positive effects (reduced cost of purchasing energy due to shortfall) and negative effects (reduced direct revenue from energy sales due to hedging) [214]. The shadow price of hedging quantifies the average marginal negative impact of hedging and increases significantly as the energy shortfall percentage or reservoir initial storages decrease. Therefore, optimal decisions regarding energy shortfall percentage are a joint function of the shadow price of hedging and the penalizing price of energy shortfall. Hydropower systems can have complex topologies with cascaded reservoirs and power plants.

Stochastic formulations represent future inflows and market prices through scenario trees or Monte-Carlo samples. The way these scenarios are generated and later reduced (e.g., forward selection, moment-matching, Kantorovich distance) has a first-order impact on the resulting water values and policy recommendations. Key sources of uncertainty typically include reservoir inflows, spot-price trajectories, fuel costs and demand levels [74,85]. Deterministic models that omit such stochastic structure are therefore confined to short horizons where forecast error is limited [51,74]. Best practice is to report the scenario-generation procedure in detail and perform robustness checks [209]. Common stress tests compare alternative statistical fits for critical drivers. For example, marginal-cost curves can be approximated with linear, polynomial, spline, or piecewise-linear functions and the sensitivity of water value to each specification is evaluated [52,58]. Models are also benchmarked against historical data to ensure they reproduce realistic price volatility and serial correlation. Non-parametric resampling techniques further relax distributional assumptions and can capture higher-order moments more faithfully [177]. The transparency of scenario design and the breadth of sensitivity analysis largely determine the credibility of stochastic water-value studies [85,144].

4.12. Common pitfalls

MILP dual misuse: quoting integer-program *duals* as water values without parametric checks leads to non-economic signals [54,156].

Inconsistent time scales: passing monthly λ_t to hourly markets without reshaping for intraday head and price patterns leads to non-representative bids [81,83].

Endogeneity mismatch: combining exogenous price forecasts with endogenous market models double-counts market response. Increasing price granularity can affect model outcomes [180].

Poorly specified terminal value: $\Phi(S_{T+1}) = 0$ inflates late-season releases and depresses λ_t prematurely [51,62,154].

Scenario under-dispersion: narrow inflow/price scenarios yield overconfident, understated λ_t spreads; disclose spread diagnostics [74].

Aggregation effects: Coarse aggregation of assets or reservoirs can understate water value by smoothing over local constraints and flexibility. Finer spatial and asset resolution tends to yield higher λ_t by revealing bottlenecks and operational opportunities [33,90,112].

Temporal granularity: Shorter time steps (e.g., hourly vs. daily) often increase water value by capturing intraday arbitrage and flexibility. Care should be taken when translating water values across time scales to ensure consistent representation of intraday head and price patterns [81,83].

5. Results and discussion of the systematic review

The systematic review highlights the range of approaches to quantifying water value, spanning classical optimisation models, stochastic programming, and equilibrium-based formulations. While these methods converge on the central principle of water value as the opportunity cost of storage, their empirical estimates differ markedly depending on modelling assumptions, temporal scope, and policy and regulatory context. The regression analysis results (Tables 3–6) provide a quantitative lens through which these differences can be systematically compared. In this section, we interpret the findings, compare methodological strengths and weaknesses, and critically evaluate the limitations of the literature. We also consider broader implications for policy and operational practice.

5.1. Descriptive statistics of the literature

We identified 277 relevant studies published between 1931 and January 2026. Fig. 3 subplots a) and c) illustrate the growth of publications over time, with a notable acceleration in the past two decades. Most of the literature consists of conference papers, followed by journal articles, primarily sourced from IEEE Xplore and Scopus. Geographic coverage is concentrated in Europe, North and South America, and China, with relatively fewer contributions from the rest of Asia and Africa.

Methodologically, LP and SDDP dominate the literature, which reflects their scalability to multireservoir systems and long horizons. Fig. 3 subplot b) shows the distribution of methodological approaches across the reviewed corpus.

Table 3
Quantitative analysis of water values, N = 462.

Dummies	(A) Logged water value	(B) Logged water value	(C) Logged water value
Econ-Engi Model	−2.02**	−0.80	−2.62**
	(1.03)	(1.16)	(1.29)
MILP Model	14.06***	11.73**	11.85***
	(1.49)	(1.52)	(1.49)
SDP Model	−4.00***	−6.33***	-6.21**
	(1.32)	(1.37)	(1.34)
SDDP Model	−4.33**	−4.46**	−4.61*
	(1.27)	(1.24)	(1.34)
Other Model	1.19	1.42	2.34*
	(1.27)	(1.29)	(1.35)
Asia-Pacific		2.50	0.40
		(2.57)	(2.59)
Europe		5.98**	4.45*
		(2.53)	(2.63)
North America		1.60	1.53
		(2.69)	(2.69)
South America		1.03	−0.23
		(2.77)	(2.73)
Environmental			8.95***
			(1.95)
Hydropower			2.34*
			(1.26)
Mixed Purpose			3.45***
			(1.15)
Urban Use			4.57***
			(1.47)
Constant	0.11	−3.54	−4.47*
	(0.09)	(2.64)	(2.62)
Adjusted R-squared	0.30	0.34	0.38

Standard errors in parentheses. *, **, *** denote significance at 10%, 5%, 1% respectively. Each water value observation is weighted by the inverse of the number of water values extracted in the original paper. Reference levels: model = LP model; region = Africa; primary use = Agricultural.

Table 4
Wald tests of coefficient equalities, based on Table 3 column (C).

Null hypothesis	F-statistics	P-value	Test result
LP = Econ-Engi = MILP = SDP = SDDP = Other	F(5, 419) = 38.07	0.00	Reject
LP = Econ-Engi = MILP = SDP = SDDP	F(4, 419) = 44.65	0.00	Reject
LP = MILP = SDP = SDDP	F(3, 419) = 58.94	0.00	Reject
LP = Econ-Engi = Other	F(2, 419) = 6.32	0.02	Reject
SDP = SDDP	F(1, 419) = 1.34	0.25	Fail to reject

Table 5
Quantitative analysis of water value uncertainty, N = 160.

Dumies	(A) Logged water value spread	(B) Logged water value spread	(C) Logged water value spread
Econ-Engi Model	−0.24	0.08	0.13
	(0.68)	(0.72)	(0.83)
MILP Model	0.79	2.09**	2.17**
	(0.97)	(0.93)	(0.90)
SDP Model	−0.19	1.11	1.20*
	(0.76)	(0.75)	(0.73)
SDDP Model	−0.58	1.22	−2.57**
	(0.79)	(0.76)	(0.85)
Other Model	0.35	−0.18	-0.09
	(0.85)	(0.88)	(0.89)
Asia-Pacific		−0.60	0.00
		(1.58)	(1.72)
Europe		−3.62**	−3.17**
		(1.42)	(1.58)
North America		−2.52	−3.52*****
		(1.58)	(1.64)
South America		0.08	0.08
		(1.50)	(1.52)
Environmental			0.49
			(1.73)
Hydropower			−0.44
			(0.99)
Mixed Purpose			1.99**
			(0.99)
Urban Use			3.53
			(2.21)
Constant	−0.80	1.52	1.43
	(0.54)	(1.49)	(1.46)
R-squared	0.02	0.14	0.20

Standard errors in parentheses. *, **, *** denote significance at 10%, 5%, 1% respectively. Each water value observation is weighted by the inverse of the number of water values extracted in the original paper. Reference levels: model = LP model; region = Africa; primary use = Agricultural.

Table 6
Wald tests of coefficient equalities, based on Table 5 column (C).

Null hypothesis	F-statistics	P-value	Test result
LP = Econ-Engi = MILP = SDP = SDDP = Other	F(5, 146) = 4.81	0.00	Reject
LP = Econ-Engi = MILP = SDP = SDDP	F(4, 146) = 5.97	0.00	Reject
LP = MILP = SDP = SDDP	F(3, 146) = 7.95	0.01	Reject
LP = Econ-Engi = Other	F(2, 146) = 0.02	0.98	Fail to reject
SDP = SDDP	F(1, 146) = 17.60	0.00	Reject

5.2. Quantitative meta-analysis of water values

As discussed in the previous sections, water values provide crucial guidance for hydroelectric dam operations, including multipurpose release schedules, market bidding strategies, and policy design. Uncertainty in future hydrological conditions and market processes naturally propagates to water value estimates, and incorporating these uncertainties is essential for a scientifically sound planning process. Motivated by these considerations, we conduct quantitative analyses of the water values surveyed from the literature to shed light on the association of different factors with water value point estimates and uncertainties. We estimate the following log-linear regression model:

$$Ln(y_i) = \beta_0 + \beta_1 \mathbb{1}\{y_i \in \mathcal{M}\} + \beta_2 \mathbb{1}\{y_i \in \mathcal{P}\} + \beta_3 \mathbb{1}\{y_i \in \mathcal{F}\} + \epsilon_i \quad (23)$$

where the dependent variable y_i represents either the water value itself or the reported range of water values from a given study, considered in the surveyed papers. Each observation is weighted by the inverse of the number of observations reported in the corresponding paper. We elect to use a log-linear functional form, as in Eq. (23), due to the nature of the water values reported in the existing literature. Namely, the water value estimates extracted from the literature spanned from 1931 to 2026, across the entire globe, and multiple distinct use purposes. The intrinsic nature of these estimates leads to a wide spread of the water value distribution. As an alternative, the linear-linear functional form is sensitive to outliers and implicitly assigns excessive weight to water values for urban use in a developed economy. The logarithmic transformation of the outcome variables compresses the distribution of water values, so that each individual estimate remains in a similar order of magnitude.

Table 3 reports regression estimates of logged water values (N = 462). Table 5 presents regression estimates of the relative range of water values, calculated as the range divided by the central estimate for each use purpose reported in the surveyed paper (N = 160). For example, if a paper examines three distinct use purposes, it contributes three separate relative range observations to the analysis. We choose this measure to evaluate the heterogeneous uncertainties embedded in water values, which may be correlated with the model's precision, water

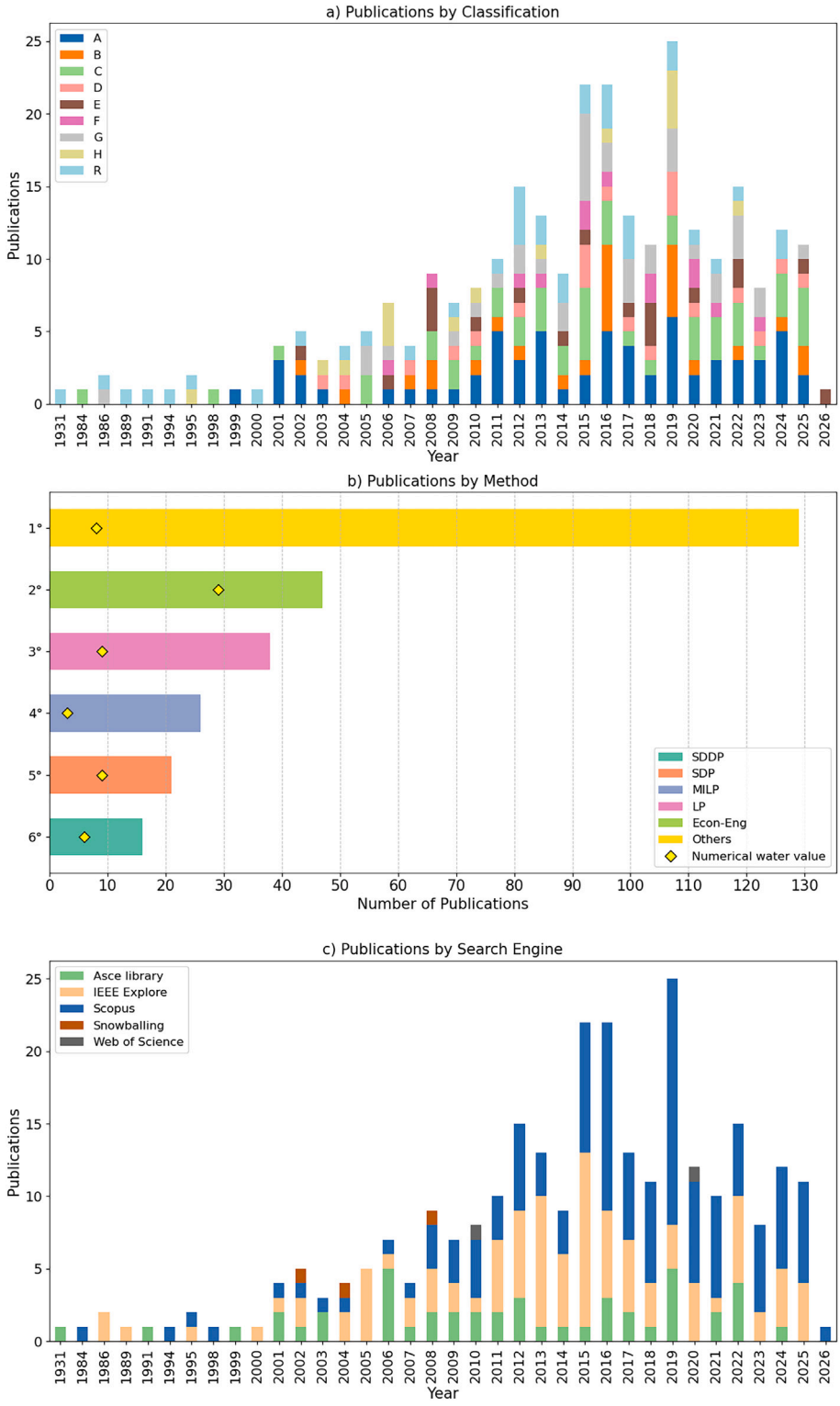


Fig. 3. Summary of the research trend in water value assessment.

use, and location. This measure serves as a proxy for the formal statistical quantity, the coefficient of variation. The coefficient of variation cannot be reliably calculated for our analysis because most studies do not provide sufficient numbers of water values for each water use purpose to allow meaningful standard deviation calculations. In both tables, Columns (A) through (C) sequentially introduce region indicators and water-use purpose indicators. Three salient patterns emerge:

(1) **Water value differences:** Relative to the baseline LP models, estimates from SDP, SDDP, and Econ–Engi models are substantially lower. By contrast, estimates from the MILP and other models are consistently higher regardless of whether regional and water-use purpose indicators are accounted for. With respect to uncertainty, MILP and SDP models tend to produce a wider spread of water value estimates than LP models.

Econ–Engi and other models yield a similar spread of water value estimates relative to the estimates from LP model.

(2) **Geographic variation:** The point estimates suggest that water values in Europe are substantially higher than those in Africa. We did not find evidence that water value estimates from other continents are statistically different than those in Africa. The differences among water value estimates may reflect hydrological abundance or policy and market factors (e.g., market design). For water value uncertainty, existing estimates exhibit regional divergence: European and North American values are relatively concentrated and more precisely estimated than the rest of the world. This disparity likely reflects distinct hydro-climatic conditions, such as the volatile inflows of tropical and Andean basins, and differing market frameworks, notably the dominant role of hydropower in Brazil's pool-based electricity market. Another potential cause of our statistical finding is that European and North American countries tend to have publicly available data with better quality, allowing for more precise estimates.

(3) **Purpose of water use:** Reservoirs with mixed purposes or those oriented toward urban and mixed supply exhibit much higher water values with high statistical precision, highlighting the premium associated with multipurpose infrastructure and critical supply systems.

The explanatory power improves substantially from Model (A) (R^2 = 0.30) to Model (C) (R^2 = 0.38). It highlights the importance of accounting for both regional context and reservoir purpose. Even when the same estimation methodology is applied, water value estimates can differ substantially across geographical regions and use purposes. Such differences are likely driven by variation in market design, resource endowments, and broader economic structures, highlighting that water value is inherently context- and time-dependent.

The higher water values observed for multipurpose and urban-supply reservoirs reflect several reinforcing mechanisms: competing non-energy uses (irrigation, municipal supply) reduce the effective storage available for power generation, which consequently increases the scarcity rent; diversion requirements can lower the effective hydraulic head; seasonal security-of-supply constraints impose minimum storage targets that further restrict operational flexibility; and regulatory penalties for supply shortfalls create additional opportunity costs that are internalised in the water value [23,87,88]. The underlying data, including all 462 water value observations, unit conversions, and analytical metadata, are documented in the companion dataset [100] and archived at the [99].

5.3. Model comparisons via Wald tests

We formally test for coefficient differences in our quantitative analysis. Tables 4 and 6 report Wald tests of the equality of methodology indicators. Specifically, we test whether all or a subset of the elements in β_3 of Eq. (23) are equal, using Wald tests and their resulting F-statistics.

We find evidence supporting the claim that methodological choice significantly influences water value estimates. As shown in Table 4, the null hypothesis that all models yield equal water values is rejected with high statistical confidence. Similarly, differences among different optimisation models, including LP, MILP, SDP, and SDDP models, are also statistically significant. For LP, Econ–Engi, MILP, and other models that are deterministic in nature, we still reject the hypothesis that they produce statistically similar water value estimates.

Results from stochastic models such as SDP and SDDP are statistically identical and lower than those from the deterministic models, suggesting that neglecting stochastic inflow and price uncertainty in deterministic formulations may lead to systematic overestimation of water values. This finding is consistent with the stochastic models' ability to internalise inflow uncertainty and hedge against low-inflow events, which depresses the expected scarcity rent relative to deterministic formulations.

Table 6 reports analogous tests for the range of water values, conditional on the same geographical region and use purpose. We reject the null hypothesis that all models generate similar ranges of water values with high statistical confidence. Subset tests on β_3 further indicate that SDDP models yield statistically significantly smaller ranges of water values compared to all other models, including SDP. In contrast, deterministic models such as LP and Econ–Engi yield water value estimates with similar ranges. This finding is somewhat counterintuitive because stochastic methods provide a more realistic representation of uncertainty; one might expect them to introduce greater volatility, and thus wider ranges in water values.

5.4. Applications in operation, planning and policy analysis

Our meta-analysis reveals that the range of water values is substantial, which reflects differences in market design, natural water endowments, and methodological approaches. These findings, synthesized from the literature, have several important implications for practice.

First, dam operations and bidding strategies should not rely on point estimates alone but instead incorporate the expected range of water values, with robustness checks against uncertainty [9,138,300]. In long-term scheduling, water value guides seasonal allocation, with stochastic models better capturing dry-year contingencies [11,36,301]. In short-term planning, deterministic LP models are still widely used due to simplicity, though they may misrepresent scarcity under uncertainty [6,39,91].

Water value can serve as the implicit bid price, aligning internal scarcity with external price signals [164,208]. To achieve an economically optimal market outcome, a dam operator must balance the marginal value of water across time periods, given a fixed price formation process [8,9]. However, this balance is complicated by the intrinsic uncertainties in water values, which stem from both uncertain inflow conditions and future market processes. A practical approach is to plan dam operations and market bidding decisions based on the expected water value, while conducting robustness checks using statistical confidence intervals around the estimated water value [37,301].

In settings where strategic bidding and production behaviours are permitted, an additional interaction arises between market prices and water values. In particular, dam operators may co-optimise by accounting for the joint impact of hydroelectric production on both market prices and water values across time periods [281]. Such interactions are especially relevant in oligopolistic markets, where the strategic influence of individual operators is more pronounced [229,230].

Second, market design and policy frameworks must recognise that hydroelectric systems are heterogeneous: multipurpose and urban-oriented reservoirs [11,34], for example, command higher water values [36,164], while regional hydrological abundance can depress them [302]. Market design studies include unit commitment transitions [201], market power mitigation [153], competition concerns [183], regulatory frameworks [161,213], renewable integration challenges [207], generator profitability [210], environmental constraint costs [240], tradable certificates [233], forward contracts [190], and hydro-dominated operations [212]. Accordingly, operational rules and market mechanisms should be calibrated to local conditions, rather than generalised from other regions or time periods [303,304]. Only by integrating use purposes and accounting for modelling choices on a case-by-case basis can scientifically sound and context-sensitive operation policies be developed [272,283]. In particular, a smaller set of studies integrates ecological or policy constraints. Here, water value acquires a multi-dimensional interpretation: both scarcity rent for power generation and the shadow price of environmental flow requirements [294]. While empirical coverage is limited, this strand highlights the versatility of the concept in sustainability-oriented water governance [10].

Third, regulatory constraints modify water-value signals in predictable ways and therefore deserve explicit attention in both

operational practice and market design. Empirical studies show that binding minimum environmental flow requirements raise the marginal value of stored water, especially in wet periods, because mandated releases lower the effective head and shorten the window for profitable generation [10]. In contrast, tight ramp-rate limits generally depress water values by curbing the plant's ability to exploit short-lived price spikes [229]. Market mechanisms that remunerate fast flexibility or allow inter-temporal trading of environmental obligations can partially offset these effects and align private scheduling incentives with broader system and ecological objectives [305].

5.5. Evolving role of water value

The meta-analysis confirms that methodological progress has materially reshaped the quantification of water value. Early cost-minimising, deterministic formulations, typified by LP and Econ–Engi models, yield lower and more stable estimates because they optimise against a single trajectory and adopt a narrow economic frame [62]. In contrast, state-of-the-art stochastic approaches such as SDP and SDDP explicitly embed inflow and price uncertainty, producing higher central estimates and wider spreads that better reflect risk and inter-temporal trade-offs [64,81]. The Wald tests reported in Table 4 reject the null of equal coefficients across modelling families, showing that the choice of optimisation technique is statistically consequential. Power market models for the clean energy transition are surveyed in [287], and market coupling consequences are explored in [178].

The literature now treats water value as a dynamic, context-dependent parameter linking plant operation with market interaction. Multi-market studies show systematically higher water values when ancillary service revenues are considered, especially under tight flexibility constraints and volatile reserve prices [70]. Likewise, when producers exercise market power, optimal bidding requires exploring a distribution of plausible water values; assuming price-taking behaviour can overstate the opportunity cost of storage and distort spillage decisions. Positive spread coefficients for SDP and SDDP models (Table 5) are consistent with this strategic use, indicating that water value has evolved from a static shadow price into a behavioural signal for risk management and competitive positioning.

5.6. Price reaction functions and hydro as price influencers

An important extension of the water value concept lies in its role within *price reaction functions*. Price reaction studies include market splitting [223], renewable penetration effects [174,182], market value of renewables [181], wholesale market power [111,243], pricing in stochastic networks [238], and alternative clearing mechanisms [140]. Large hydropower systems and PSH units are often *price influencers* in their respective markets: their release and storage decisions directly shift supply, affecting Locational Marginal Prices (LMPs) and ancillary service prices.

Despite this inherent market influence, many traditional operational and hydrological models treat prices as exogenous variables, omitting the feedback loops between hydro dispatch decisions and market outcomes [78]. Such simplified modelling can produce systematic distortions and suboptimal allocation policies, particularly in markets with high hydro penetration and limited ancillary-service depth [107]. Our regression and meta-analysis support this critique: models that explicitly incorporate strategic price response (through equilibrium or supply-function formulations) tend to yield higher estimates of water value, consistent with observed historical market behaviour. By contrast, purely optimisation-based models (LP, SDP, SDDP) often underrepresent this interaction and compress the estimated scarcity rents.

Methodologically, price-responsive behaviour can be modelled in several complementary ways.

(i) Econometric or reduced-form price models estimate how market prices respond to exogenous demand shifters and fuel costs, providing an empirical supply curve that strategic hydro producers can take as given [67].
(ii) Full equilibrium or market-clearing formulations couple strategic offers with an explicit supply–demand balance, using game-theoretic or bi-level optimisation techniques to capture feedback between bids and clearing prices [33].
(iii) A computationally lighter variant fixes the aggregate offer of all *non-strategic* units and lets the strategic hydro producer face the resulting *residual demand* curve; in effect, the producer treats its own dispatch decision as a demand shifter against a static residual supply, avoiding the need to model other agents' strategic reactions explicitly. This residual-demand approach preserves the core incentive structure while sidestepping the dimensionality of a full equilibrium.
(iv) Machine-learning surrogates learn non-linear price responses directly from historical data but may sacrifice interpretability and require large training sets [40]. In large integrated energy-system models, any of the above representations can become computationally burdensome. Aggregation of hydropower plants or temporal resolution is therefore often employed to maintain tractability while keeping the approximation error within acceptable limits [90].

Future progress will likely hinge on hybrid approaches that blend structural optimisation with empirical calibration against observed price reaction functions. For example, long-term price forecasting in hydro-thermal power systems can utilise a framework consisting of a stochastic long-term strategic model to provide water values to a more detailed short-term operational model [206]. The ability of econometric models to verify established views on hydropower supply drivers and assess the relative importance of various factors, coupled with their predictive capacity, makes them valuable for both market participants and energy market modellers [68].

5.7. Limitations in the literature

Despite these insights, the regressions also reveal persistent limitations. The relatively low R^2 in the base models (Table 3, column A) indicates that methodological dummies alone do not explain most of the variation in water values. This points to the importance of unreported assumptions, such as head–efficiency curves, scenario design, or terminal values, in shaping results. Moreover, the lack of standardised reporting across studies complicates meta-analysis and comparability, as many coefficients may reflect the influence of omitted variables. Accordingly, the meta-analytic results should be interpreted descriptively.

The spread regressions further indicate the lack of robustness: large dispersion in λ_t estimates suggests sensitivity not just to method, but also to modelling choices rarely disclosed. Moreover, environmental objectives are underrepresented, as few studies explicitly co-optimise ecological releases. This omission is striking given the importance of environmental trade-offs in contemporary hydropower governance.

5.8. Recommendations for temporal and computational performance

Another major challenge lies in balancing model accuracy with tractable solution times. Our review and regression evidence highlight that SDDP-based approaches provide strong treatment of uncertainty but at significant computational cost, while equilibrium formulations suffer from scalability bottlenecks when extended to multireservoir or multi-agent systems. Improving temporal performance and computational efficiency requires several steps:

(i) **Temporal granularity:** adopt rolling-horizon formulations and multi-resolution time steps (e.g., hourly short-term operation nested within seasonal planning).

(ii) **Computational efficiency:** employ parallelisation, decomposition, and warm-start techniques to reduce solution times, especially for large SDDP instances.
(iii) **Accuracy and reliability assessment:** systematically report out-of-sample validation of water value policies, including stress-testing against hydrological extremes and price shocks.
(iv) **Trade-off quantification:** quantify explicitly the balance between runtime and solution quality (e.g., bound gaps in SDDP, convergence speed in equilibrium models).

Embedding these practices would both improve comparability across studies and ensure robustness of results for market and policy applications.

5.9. Policy and operational implications

Our synthesis indicates that water value is both critical to hydropower decision-making and highly context dependent. The quantitative evidence shows that estimated water values and their dispersion vary systematically with modelling approach and region, with stochastic formulations (SDP/SDDP) yielding wider spreads than purely economic–engineering models, and with notable differences across South America, North America, and Europe [106,112,144,306]. These patterns have direct implications for policy and operations: they caution against the use of point estimates outside their original context, account explicitly for uncertainty, and call for clearer documentation of modelling assumptions that influence the shadow price of storage (e.g., inflow scenario design, head–efficiency representation, and treatment of co-optimised products) [70,74,75,84].

For regulators, an immediate priority is to improve transparency and comparability in how water values are produced and used in market participation. We recommend hydropower participants to document the methodological provenance of water values embedded in bids, including the objective function (profit vs. welfare), the salvage value function and its calibration, the stochastic structure and coverage of hydrology and prices, and the handling of head effects and non-convexities [61,106]. Given the higher volatility associated with uncertainty-aware models, market rules should acknowledge and accommodate risk explicitly, for example, by permitting or encouraging mean risk bidding policies grounded in coherent risk measures (such as CVaR [307]), and by stress-testing bidding strategies against hydrological extremes [54,214]. Because hydropower can be a price influencer in several markets, monitoring protocols and market power screens should be adjusted to hydro-dominated systems, where intertemporal opportunity costs and reservoir constraints shape strategic behaviour differently than for thermal fleets [18,51]. Finally, regional calibration is critical given the documented heterogeneity in scarcity rents across basins and market structures and rules; uniform prescriptions are unlikely to be efficient. Regulators should require periodic benchmarking and out-of-sample validation of operators' water value policies against local hydrology and market conditions [55,75,122].

For market designers, the evidence favours mechanisms that internalise intertemporal opportunity costs across all relevant products. Co-optimisation of energy and ancillary services should reflect the asymmetric water implications of reserve provision, such as the increased water burn required to maintain down-regulation capacity. Settlement rules and product definitions need to align with physically meaningful activation probabilities and ramping constraints, so that the marginal revenue of reserves is proportional to the induced change in the water value. PSH arbitrage conditions should be modelled and settled consistently across day-ahead and real-time horizons to avoid distortions in implied water valuations [43,78,81]. Given the persistence of uncertainty in both hydrology and prices, the development of hydro-relevant hedging instruments and forward products—designed with transparent links to the underlying physical scarcity rent—can improve risk sharing and investment signals [107]. Market data access sufficient to reconstruct and validate water value trajectories ex post would materially improve learning and oversight [18,61].

For dam operators, operational practice should move beyond deterministic point estimates toward policies that explicitly account for the distribution of water values over relevant horizons. In practical terms, this implies adopting stochastic scheduling frameworks with rolling horizons and multi-resolution time steps; using confidence bands around water value estimates to inform bidding and release decisions; and subjecting policies to systematic out-of-sample tests under dry, median, and wet sequences [74,85,253]. Particular attention should be paid to modelling choices that can systematically alter short-term water values, including the specification of terminal value functions, aggregation across cascades, and the conversion between volumetric and energy units under varying head [12,35]. Co-optimisation with ancillary services should be evaluated with plant-specific constraints to avoid inadvertent overconsumption of water for reserve commitments. Investments in forecasting, telemetry, and flexible turbine operation will increase the value of information and the ability to monetise intraday and seasonal arbitrage without compromising long-term storage objectives [13,144].

6. Future research directions

While the current body of research provides important insights, several frontiers remain open. We highlight four directions that, if pursued, could substantially advance both academic understanding and practical utility.

6.1. Integrating uncertainty and environmental objectives

Future work should develop models that co-optimise energy and ecological objectives under uncertainty and consider ecosystem valuation [195,226,256], hydro-solar complementarity [225], integrated wind-PV-hydro systems [159], environmental demand reallocation [245], and water-energy-food nexus models [131]. Stochastic programming frameworks, especially SDDP, provide a natural basis for coupling these objectives, but extensions are needed to capture ecological state variables (e.g., habitat connectivity, temperature dynamics). Such models would allow water values to reflect trade-offs not just between present and future energy revenues, but also between economic and environmental goals.

6.2. Coupling physical and financial water value

As electricity markets mature, the translation of physical water values into financial products (e.g., forward contracts, derivatives [266,267], and Financial Transmission Rights [268]) becomes increasingly relevant. Research is needed to explore how reservoir shadow prices interact with day-ahead and real-time market settlement rules, particularly in the presence of reserves, ancillary services, and transmission constraints. Such coupling could provide hedging tools for hydropower producers and enhance market transparency [200].

6.3. Role of hydrological variability

Long-term shifts in hydrological patterns are reshaping inflow regimes and altering the statistical properties [109,173], on which stochastic optimisation models rely [7]. Few studies explicitly test the robustness of water values under shifting hydrological baselines, increased volatility, or non-stationary inflows [16,55,73]. Future research should assess how environmentally induced hydrological shifts propagate into water values, operational strategies, and investment signals. This would inform adaptive reservoir management and market design.

6.4. Interdisciplinary extensions

Finally, advancing the concept of water value requires deeper interdisciplinary engagement. Interdisciplinary extensions include water

district organization [257], water sharing frameworks [204,262], water privatization limits [205], sustainable water resources [261], urban water management [221,227], agent-hydrologic models [248], groundwater economics [237,241,255,269], water market clearinghouses [234], economy-ecology conversion [250], hydropower relicensing [254], reservoir sustainability [275], social costs of transmission [242], and marginal cost pricing history [276]. Legal scholarship can clarify how water rights and property regimes affect the distribution of water values. Ecological research can provide metrics for valuing non-market services. Political science can illuminate how institutions and governance structures mediate trade-offs across stakeholders. Only by bridging these disciplines can water value evolve from a narrow economic construct to a detailed tool for sustainable river basin management. This comprehensive approach must further incorporate the crucial water-energy-food nexus, which recognises that operational decisions and resulting water value are intrinsically linked to competing demands such as agricultural irrigation, human consumption, and multi-use constraints on stored water.

7. Conclusion

This review clarifies water value as the marginal opportunity cost of stored water, the shadow price of the reservoir balance, and distinguishes it from marginal production cost and market-clearing price. We assemble a unified analytical framework spanning LP/MILP, SDP/SDDP, and economic–engineering models, and show how hydrologic scarcity, market conditions (including co-optimised reserves), and operational constraints jointly shape water values. Stochastic formulations (SDP/SDDP) better internalise uncertainty and hedging behaviour, while MILP captures non-convex operating detail; each entails characteristic trade-offs in physical fidelity, interpretability, and scalability.

The quantitative synthesis indicates that methodological choice materially affects both levels and dispersion of estimated water values: stochastic models tend to yield lower central values and wider spreads relative to deterministic LP, whereas MILP often produces higher central values once operational detail is represented. Wald tests confirm these differences. Values are highly context-dependent, varying with region and reservoir purpose (e.g., multipurpose and urban supply systems exhibit higher values). These findings support moving beyond point estimates toward risk-aware policies that report assumptions (salvage value, scenario design, head effects) and account for reserve asymmetries and PSH arbitrage.

For practice and policy, we recommend: adopting stochastic scheduling with rolling horizons and consistent cross-scale modelling in cascades; co-optimising energy and ancillary services with physically meaningful activation and ramping; and improving transparency in the provenance of water values used in bidding. Priority research gaps include integrating environmental objectives under uncertainty, addressing non-stationary hydrology, coupling physical water values to financial risk management, and standardising reporting to improve reproducibility and comparability.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Argo, the official Argonne National Laboratory (ANL) AI assistant, to assist with grammar and language refinement. The authors subsequently reviewed and edited the content as needed and assume full responsibility for the final version of the publication.

Funding

This work was authored by Argonne National Laboratory, operating under contract number DE-AC02-06CH11357, with the U.S. Department of Energy (DOE) under the Hydropower and Hydrokinetic Office (H2O). Funding was provided through the H2O Seedling Program, which supports early-stage, high-risk, high-reward research in water power technologies.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Matija Pavicevic reports that financial support was provided by the U.S. Department of Energy Water Power Technologies Office. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Terminology comparison table

Table A.7 provides a detailed mapping of the principal terms used across research communities to describe the marginal value of stored water. All terms refer to the same underlying concept; this paper standardises on “water value” throughout.

Table A.7
Terminology mapping for water value concepts across research communities.

Term	Definition (as used in this paper)	Typical sub-community	Usage context	Ref.
Marginal water value	Expected future benefit of storing one additional unit of water; the derivative of the value function with respect to storage.	Operations Research, Hydro-economics	Reservoir scheduling, medium- and long-term planning	[19,64]
Opportunity cost of water	Value of the best alternative use foregone by allocating water to its current use (e.g., power generation).	Electricity-market design, Hydro-economics	Market bidding, short-term dispatch	[20,22]
Scarcity rent	Premium generated by water scarcity; the difference between the marginal benefit of water and its marginal extraction or conveyance cost.	Hydro-economics, Resource economics	Basin-wide allocation, water policy	[23,24]
Shadow price / Shadow rent	Lagrange multiplier on the reservoir water mass balance constraint in an optimisation model.	Operations Research	Deterministic and stochastic optimisation models	[26,28]
Dual value / Dual price	Change in the objective function value per unit relaxation of the storage balance constraint; obtained from LP or MILP duality theory.	Operations Research	LP/MILP scheduling, sensitivity analysis	[29,30]

Appendix B. Methods vs. use-case decision guide

Table B.8
Methods vs. use-case decision guide: indicative suitability of each methodology under different problem characteristics.

Problem characteristic	LP	MILP	SDP	SDDP	Econ–Engi	Other
Small cascade (1–3 reservoirs)	High	High	High	Medium	High	Variable
Large cascade (>10 reservoirs)	High	Medium	Low	High	Medium	Low
High inflow volatility	Low	Low	High	High	Medium	Variable
Day-ahead bidding	High	High	Medium	Medium	Low	Medium
Ancillary service co-optimisation	Medium	High	Medium	Medium	Low	Low
Pumped-storage presence	Medium	High	Medium	Medium	Low	Medium
Ecological constraints	Low	Medium	High	Medium	High	Variable
Long-term planning (>1 year)	Medium	Low	Medium	High	High	Variable

Note: Suitability ratings reflect the general consensus from the reviewed literature. "High" indicates the method is well suited and widely applied; "Medium" indicates the method is applicable with caveats or additional effort; "Low" indicates significant limitations or rare application; "Variable" indicates suitability depends heavily on the specific implementation or variant used.

Table B.8 provides practitioners with indicative guidance on the suitability of each major methodology under different operational conditions and problem characteristics. Suitability ratings (High, Medium, Low, Variable) reflect the consensus emerging from the reviewed literature and the comparative analysis in Section 4.11; they should be interpreted as starting points rather than definitive prescriptions, since site-specific factors (data availability, computational budget, regulatory requirements) will influence the final choice.

Appendix C. Extended analytical derivations

This appendix provides detailed mathematical derivations and extended formulations that underpin the summary models and concepts presented in the main text. It is intended for readers seeking additional rigor, implementation guidance, or deeper insight into the mechanics of water value estimation in hydropower and multi-sectoral water systems.

C.1. Water demand framework details

Motivation: The water demand framework models the optimal allocation of a limited water resource among competing uses, capturing the marginal value of water as a shadow price. Here, we provide the full Lagrangian and first-order conditions.

Suppose there is a limited water resource $Q > 0$ and p potential uses. Let $q_p \geq 0$ denote the amount of water allocated to use p. Each use p yields a (gross) benefit $U_p(q_p)$, with $U_p'(q_p) > 0$ and $U_p''(q_p) < 0$.

The social planner's problem is:

$$\max_{\{q_p\}_{p\in\mathcal{P}}} \sum_{p\in\mathcal{P}} U_p(q_p) \quad \text{s.t.} \quad \sum_{p\in\mathcal{P}} q_p \leq Q, \quad q_p \geq 0 \tag{C.1}$$

The Lagrangian is:

$$\mathcal{L}(q,\lambda,\mu) = \sum_{p\in\mathcal{P}} U_p(q_p) - \lambda\left(\sum_{p\in\mathcal{P}} q_p - Q\right) - \sum_{p\in\mathcal{P}} \mu_p q_p \tag{C.2}$$

First-order KKT conditions:

$$U_p'(q_p^\star) - \lambda^\star - \mu_p^\star = 0 \quad (p\in\mathcal{P}) \tag{C.3}$$

$$\lambda^\star\left(\sum_{p=1}^{m} q_p^\star - Q\right) = 0, \quad \mu_p^\star q_p^\star = 0 \tag{C.4}$$

$$\lambda^\star \geq 0, \quad \mu_p^\star \geq 0, \quad q_p^\star \geq 0, \quad \sum_{p=1}^{m} q_p^\star \leq Q \tag{C.5}$$

For any use with $q_p^\star > 0$, $U_p'(q_p^\star) = \lambda^\star$. Thus, the shadow price $\lambda^\star$ is the marginal value of water, equalized across all active uses.

Critical notes:

- The approach is sensitive to the specification of $U_p(\cdot)$ and the accuracy of demand data.
- In practice, physical, legal, and regulatory constraints may further restrict allocations.

C.2. Cascaded reservoir systems and propagation of water values

Motivation: In multireservoir river systems, water value propagation and upstream-downstream interactions are complex. This section details the mathematical formulation for cascaded systems, including routing, Lagrangian construction, and sensitivity analysis.

C.2.1. Notation and routing

Index reservoirs by $r \in \mathcal{R}$. Let $\mathcal{U}(r)$ denote the set of immediate upstream reservoirs that release into reservoir r; let $\mathcal{D}(r)$ denote the immediate downstream reservoir(s) or reaches receiving flow from r. We allow for travel time delays $\tau_{u\to r} \in \mathbb{Z}_{\geq 0}$ (in time-steps) and fractional losses/partitioning $\gamma_{u\to r} \in [0,1]$ (accounting for routing losses, diversions, or portion routed to r). Upstream releases therefore affect downstream inflows with a convolutional structure.

The multireservoir storage balance for reservoir r is written:

$$\begin{aligned} S_{t+1}^{(r)} = S_t^{(r)} + I_t^{(r)} + \sum_{u\in\mathcal{U}(r)} \gamma_{u\to r}\, R_{t-\tau_{u\to r}}^{\mathrm{pow},(u)} \\ + Q_t^{\mathrm{pump},(r)} - R_t^{\mathrm{pow},(r)} - R_t^{\mathrm{npr},(r)} - \mathrm{Sp}_t^{(r)} - E_t^{(r)} \end{aligned} \tag{C.6}$$

with $S^{(r),\min} \leq S_t^{(r)} \leq S^{(r),\max}$. All time indices outside the horizon should be interpreted via appropriate boundary conditions (e.g., zero or exogenous).

Power conversion at reservoir r uses the usual conversion factor $\kappa_t^{(r)}$ (volume-to-energy; see Eq. (3) in the main text):

$$G_t^{(r)} = \kappa_t^{(r)}\, R_t^{\mathrm{pow},(r)}. \tag{C.7}$$

C.2.2. System objective and Lagrangian

Consider a (convex) welfare or cost-minimisation formulation that includes hydropower revenues (or avoided costs) and operating costs for other generators. For simplicity write the instantaneous objective as $\Pi_t(\cdot)$ and the total objective as $\sum_{t=0}^{T}\Pi_t$. Form the Lagrangian by introducing multipliers $\lambda_t^{(r)}$ for each reservoir balance (C.6) (we keep the sign convention consistent with the main text—λ multiplies the equality

written in the form $S_{t+1} - S_t - \cdots = 0$).

$$\mathcal{L} = \sum_{t=0}^{T} \Bigg\{ \Pi_t + \sum_{r\in\mathcal{R}} \lambda_t^{(r)} \Big(S_{t+1}^{(r)} - S_t^{(r)} - I_t^{(r)} - \sum_{u\in U^*(r)} \gamma_{u\to r}\, R_{t-\tau_{u\to r}}^{\mathrm{pow},(u)} + R_t^{\mathrm{pow},(r)} + R_t^{\mathrm{npr},(r)} + \mathrm{Sp}_t^{(r)} + E_t^{(r)} - Q_t^{\mathrm{pump},(r)} \Big) \Bigg\} + (\text{other multipliers for generation limits, transmission, etc.}). \tag{C.8}$$

C.2.3. First-order stationarity for upstream releases (interpretation)

Take a partial derivative of $\mathcal{L}$ with respect to the upstream *power release* variable $R_t^{\mathrm{pow},(u)}$ for some reservoir u. We assume an interior optimum (no active saturating bound on that variable); boundary cases add complementary terms. Because $R_t^{\mathrm{pow},(u)}$ contributes to the instantaneous objective through energy $G_t^{(u)}$ and enters the Lagrangian also via the local reservoir u and via downstream reservoir balances (after travel delays), the derivative collects three types of terms:

- **Immediate revenue (or marginal contribution to objective):** $\partial \Pi_t / \partial R_t^{\mathrm{pow},(u)} = p_t\, \kappa_t^{(u)}$ (for energy revenue at nodal price p_t, or an appropriate marginal cost substitute in cost minimisation).
- **Local storage effect:** differentiation of the term $\lambda_t^{(u)} R_t^{\mathrm{pow},(u)}$ in (C.8) contributes $+\lambda_t^{(u)}$.
- **Downstream coupling:** $R_t^{\mathrm{pow},(u)}$ appears (with coefficient $-\gamma_{u\to r}$) inside the balance of each downstream reservoir $r \in D(u)$ at time $t+\tau_{u\to r}$. Differentiation therefore contributes $-\sum_{r\in D(u)} \gamma_{u\to r}\, \lambda_{t+\tau_{u\to r}}^{(r)}$.

Setting the derivative to zero (stationarity) yields the key relation

$$\boxed{p_t\, \kappa_t^{(u)} + \lambda_t^{(u)} - \sum_{r\in D(u)} \gamma_{u\to r}\, \lambda_{t+\tau_{u\to r}}^{(r)} = 0} \tag{C.9}$$

Interpretation. Re-arrange (C.9) as

$$p_t \kappa_t^{(u)} = \sum_{r\in D(u)} \gamma_{u\to r} \lambda_{t+\tau_{u\to r}}^{(r)} - \lambda_t^{(u)}. \tag{C.10}$$

A unit of water released from reservoir u yields immediate revenue $p_t \kappa_t^{(u)}$, but it removes one unit of upstream storage (loss of upstream marginal value $\lambda_t^{(u)}$) and increases downstream storage (or reduces downstream generation requirements) after travel time τ, creating downstream marginal benefit $\gamma \lambda^{(r)}$. The stationarity condition equates marginal immediate revenue to the net change in (downstream benefits minus upstream opportunity cost). In words: you release if and only if immediate price compensates for the loss of upstream option value net of the downstream value created by that release.

Special case: single immediate downstream reservoir, no loss, no lag. If $D(u) = \{s\}$, $\gamma_{u\to s} = 1$, $\tau = 0$, Eq. (C.9) reduces to:

$$p_t \kappa_t^{(u)} + \lambda_t^{(u)} - \lambda_t^{(s)} = 0 \quad \Longrightarrow \quad p_t \kappa_t^{(u)} = \lambda_t^{(s)} - \lambda_t^{(u)} \tag{C.11}$$

Thus, the nodal price multiplied by the volume-to-energy factor equals the difference between downstream and upstream marginal values. This makes clear the economic logic: upstream release occurs when the downstream marginal value exceeds the upstream marginal value by at least the immediate price-adjusted revenues.

C.2.4. Chain-rule expression for systemic sensitivity

Often, one is interested in the *total* marginal value of increasing an upstream reservoir storage $S_t^{(u)}$ by a small amount, accounting for all future routing and policy adaptations. A useful formal (but not computationally trivial) representation is the following.

$$\frac{\partial V}{\partial S_t^{(u)}} = \lambda_t^{(u)} + \sum_{r\in\mathcal{R}} \sum_{\tau\geq 0} \lambda_{t+\tau}^{(r)} \frac{\partial I_{t+\tau}^{(r)}}{\partial S_t^{(u)}}, \tag{C.12}$$

where $\partial I_{t+\tau}^{(r)} / \partial S_t^{(u)}$ captures the (possibly non-linear) propagation of an upstream storage perturbation into downstream natural inflows after routing, including the effect of changed operating policies. The second term thus aggregates downstream marginal values weighted by the routing sensitivities. In practice, computing $\partial I / \partial S$ requires linearisation of routing and/or policy-response models (or simulation with finite differences).

C.2.5. Practical implications and computational notes

Travel time and discounting. Travel delays τ shift the time at which the downstream marginal values apply. If the objective is discounted, the RHS of (C.9) should include discount factors δ^{τ}.

Losses and partitioning. $\gamma_{u\to r} < 1$ reduces the effective downstream benefit per unit released upstream; diverting flows to multiple downstream nodes splits the marginal downstream benefit.

Network/diversion complexity. For large basin networks, the routing operator can be expressed by a sparse convolution matrix R such that downstream inflows $I^{\downarrow} = R\, R^{\mathrm{pow}}$ (with time indexing). Stationarity conditions become a linear system coupling all $\lambda^{(r)}$.

Policy dependence. Eq. (C.9) reflects local marginality given the current policy; if a small storage perturbation leads the optimal policy to change discretely (e.g., different release regimes), marginal approximations may be invalid, and one should compute finite-difference water values.

SDP/SDDP viewpoint. SDP and SDDP capture these propagation effects implicitly within the value function $V_t(\mathbf{S}_t)$; the slope of V with respect to each component $S^{(r)}$ therefore already reflects downstream coupling. However, SDDP's linear cuts may smear localized non-linear routing effects if stagewise-independence or linear head assumptions are used.

Appendix D. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.rser.2026.116952.

Data availability

The extracted database, classification metadata, and regression code are publicly available at https://doi.org/10.5281/zenodo.18778100 and are described in the companion Data in Brief article.

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